

MEA Compliance Test Report

MEA OCPP Certification

January 5, 2026

MEA OCPP 1.6 Compliance Verification Matrix

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
1. Configuration Verification					
1.1	CS → CSMS	BootNotification	chargePointModel, chargePointVendor, firmwareVersion	[CSMS] <- BootNotification: {...} [CSMS] -> Conf: status=Accepted	Pass
1.2	CS → CSMS	StatusNotification	connectorId, errorCode, info, status, timestamp, vendorId	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
1.3	CSMS → CS	TriggerMessage (BootNotification)	status (Accepted)	[CSMS] -> TriggerMessage: BootNotification [CS] -> CallResult: status=Accepted	Pass
1.4	CS → CSMS	BootNotification (Triggered)	chargePointModel, chargePointVendor	[CSMS] <- BootNotification: {...} [CSMS] -> Conf: status=Accepted	Pass
1.5	CSMS → CS	TriggerMessage (StatusNotification)	status (Accepted)	[CSMS] -> TriggerMessage: StatusNotification [CS] -> Conf: status=Accepted	Pass
1.6	CS → CSMS	StatusNotification (Triggered)	connectorId, status	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
1.7	CSMS → CS	TriggerMessage (MeterValues)	status (Accepted)	[CSMS] -> TriggerMessage: MeterValues [CS] -> Conf: status=Accepted	Pass
1.8	CS → CSMS	MeterValues	connectorId, meterValue	[CSMS] <- MeterValues: {...} [CSMS] -> Conf: {}	Pass
1.9	CSMS → CS	GetConfiguration	HeartbeatInterval	[CSMS] -> GetConfiguration [CS] -> Conf: HeartbeatInterval=60	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
1.10	CSMS → CS	ChangeConfiguration (Heartbeat)	status (Accepted)	[CSMS] -> ChangeConfig: Heartbeat=600 [CS] -> Conf: status=Accepted	Pass
1.11	CSMS → CS	ChangeConfiguration (MeterValue)	status (Accepted)	[CSMS] -> ChangeConfig: MeterValue=2 [CS] -> Conf: status=Accepted	Pass
1.12	CSMS → CS	ChangeConfiguration (Unlock)	status (Accepted)	[CSMS] -> UnlockConnector: 1 [CS] -> Conf: status=Unlocked	Pass
1.13	CSMS → CS	ChangeAvailability (Inoperative)	status (Accepted)	[CSMS] -> ChangeAvailability: Inoperative [CS] -> Conf: status=Accepted	Pass
1.14	CS → CSMS	StatusNotification	status: Unavailable	[CSMS] <- StatusNotification: {'status': 'Unavailable'} [CSMS] -> Conf: {}	Pass
1.15	CSMS → CS	ChangeAvailability (Operative)	status (Accepted)	[CSMS] -> ChangeAvailability: Operative [CS] -> Conf: status=Accepted	Pass
1.16	CS → CSMS	StatusNotification	status: Available	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
1.17	CSMS → CS	GetDiagnostics	fileName	[CSMS] -> GetDiagnostics [CS] -> Conf: fileName=diagnostics.log	Pass
1.18	CS → CSMS	DiagnosticsStatusNotification	status: Uploaded	[CSMS] <- DiagnosticsStatus: {'status': 'Uploaded'} [CSMS] -> Conf: {}	Pass
1.19	CSMS → CS	UpdateFirmware	location (URI)	[CSMS] -> UpdateFirmware [CS] -> Conf: {}	Pass
1.20	CS → CSMS	FirmwareStatusNotification	status: Installed	[CSMS] <- FirmwareStatus: {'status': 'Installed'} [CSMS] -> Conf: {}	Pass
1.21	CSMS → CS	ChangeConfiguration (LocalAuth)	status (Accepted)	[CSMS] -> ChangeConfig: LocalAuth=True [CS] -> Conf: status=Accepted	Pass
1.22	CSMS → CS	SendLocalList	status (Accepted)	[CSMS] -> SendLocalList [CS] -> Conf: status=Accepted	Pass
1.23	CSMS → CS	GetLocalListVersion	listVersion	[CSMS] -> GetLocalListVersion [CS] -> Conf: listVersion=1	Pass
1.24	CSMS → CS	ClearCache	status (Accepted)	[CSMS] -> ClearCache [CS] -> Conf: status=Accepted	Pass
2. Auto Charge Verification					
2.1	CSMS → CS	ChangeConfiguration (AutoCharge)	status (Accepted)	[CSMS] -> ChangeConfig: AutoCharge=False [CS] -> Conf: status=Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
2.2	CS → CSMS	StatusNotification (Plug)	status: Preparing	[CSMS] <- StatusNotification: {'status': 'Preparing'} [CSMS] -> Conf: {}	Pass
2.3	CS → CSMS	StatusNotification (Un-plug)	status: Available	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
2.4	CSMS → CS	ChangeConfiguration (Auto)	status (Accepted)	[CSMS] -> ChangeConfig: AutoCharge=True [CS] -> Conf: status=Accepted	Pass
2.5	CS → CSMS	StatusNotification (Plug)	status: Preparing	[CSMS] <- StatusNotification: {'status': 'Preparing'} [CSMS] -> Conf: {}	Pass
2.6	CS → CSMS	Authorize (Auto)	status: Accepted	[CSMS] <- Authorize: {'id_tag': 'VID...'} [CSMS] -> Conf: status=Accepted	Pass
2.7	CS → CSMS	StartTransaction	status: Accepted	[CSMS] <- StartTransaction [CSMS] -> Conf: status=Accepted	Pass
2.8	CS → CSMS	StatusNotification	status: Charging	[CSMS] <- StatusNotification: {'status': 'Charging'} [CSMS] -> Conf: {}	Pass
2.9	CS → CSMS	MeterValues	measurand	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
2.10	CSMS → CS	RemoteStopTransaction	status: Accepted	[CSMS] -> RemoteStop [CS] -> Conf: status=Accepted	Pass
2.11	CS → CSMS	StopTransaction	status: Accepted	[CSMS] <- StopTransaction: {'reason': 'Remote'} [CSMS] -> Conf: status=Accepted	Pass
2.12	CS → CSMS	StatusNotification	status: Finishing	[CSMS] <- StatusNotification: {'status': 'Finishing'} [CSMS] -> Conf: {}	Pass
2.13	CS → CSMS	StatusNotification	status: Available	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
3. Normal Operation Verification					
3.1	CS → CSMS	StatusNotification	status: Available	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
3.3	CS → CSMS	StatusNotification	status: Preparing	[CSMS] <- StatusNotification: {'status': 'Preparing'} [CSMS] -> Conf: {}	Pass
3.4	CS → CSMS	Authorize	status: Accepted	[CSMS] <- Authorize [CSMS] -> Conf: status=Accepted	Pass
3.5	CS → CSMS	StartTransaction	status: Accepted	[CSMS] <- StartTransaction [CSMS] -> Conf: status=Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
3.6	CS → CSMS	StatusNotification	status: Charging	[CSMS] <- StatusNotification: {'status': 'Charging'} [CSMS] -> Conf: {}	Pass
3.7	CS → CSMS	MeterValues	Values	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
3.8	CS → CSMS	StopTransaction (Local)	status	[CSMS] <- StopTransaction: {'reason': 'Local'} [CSMS] -> Conf: status=Accepted	Pass
3.9	CS → CSMS	StatusNotification	status: Finishing	[CSMS] <- StatusNotification: {'status': 'Finishing'} [CSMS] -> Conf: {}	Pass
3.10	CS → CSMS	StatusNotification	status: Available	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
3.11	CS → CSMS	StatusNotification	status: Preparing	[CSMS] <- StatusNotification: {'status': 'Preparing'} [CSMS] -> Conf: {}	Pass
3.12	CSMS → CS	RemoteStartTransaction	status: Accepted	[CSMS] -> RemoteStart [CS] -> Conf: status=Accepted	Pass
3.13	CS → CSMS	StartTransaction	status: Accepted	[CSMS] <- StartTransaction [CSMS] -> Conf: status=Accepted	Pass
3.14	CS → CSMS	StatusNotification	status: Charging	[CSMS] <- StatusNotification: {'status': 'Charging'} [CSMS] -> Conf: {}	Pass
3.15	CS → CSMS	MeterValues	Values	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
3.16	CSMS → CS	RemoteStopTransaction	status: Accepted	[CSMS] -> RemoteStop [CS] -> Conf: status=Accepted	Pass
3.17	CS → CSMS	StopTransaction (Remote)	status	[CSMS] <- StopTransaction: {'reason': 'Remote'} [CSMS] -> Conf: status=Accepted	Pass
3.18	CS → CSMS	StatusNotification	status: Finishing	[CSMS] <- StatusNotification: {'status': 'Finishing'} [CSMS] -> Conf: {}	Pass
3.19	CS → CSMS	StatusNotification	status: Available	[CSMS] <- StatusNotification: {'status': 'Available'} [CSMS] -> Conf: {}	Pass
4. Reset Verification					
4.1	CSMS → CS	Reset (Hard)	status (Accepted)	[CSMS] -> Reset (Hard) [CS] -> Conf: status=Accepted	Pass
4.2	CS → CSMS	BootNotification	Post-reboot checks	[CSMS] <- BootNotification [CSMS] -> Conf: status=Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
4.3	CS → CSMS	Heartbeat	Received	[CSMS] <- Heartbeat [CSMS] -> Conf: {}	Pass
4.4	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
4.5	CS → CSMS	StatusNotification	status: Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
4.6	CSMS → CS	ChangeConfiguration (Heartbeat)	status (Accepted)	[CSMS] -> ChangeConfig: Heartbeat=600 [CS] -> Conf: status=Accepted	Pass
4.7	CSMS → CS	ChangeConfiguration (MeterValue)	status (Accepted)	[CSMS] -> ChangeConfig: MeterValue=60 [CS] -> Conf: status=Accepted	Pass
4.8	CSMS → CS	ChangeConfiguration (LocalAuth)	status (Accepted)	[CSMS] -> ChangeConfig: LocalAuth=True [CS] -> Conf: status=Accepted	Pass
4.9	CSMS → CS	Reset (Soft)	status (Accepted)	[CSMS] -> Reset (Soft) [CS] -> Conf: status=Accepted	Pass
4.10	CS → CSMS	BootNotification	Post-reboot checks	[CSMS] <- BootNotification [CSMS] -> Conf: status=Accepted	Pass
4.11	CS → CSMS	Heartbeat	Received	[CSMS] <- Heartbeat [CSMS] -> Conf: {}	Pass
4.12	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
4.13	CS → CSMS	StatusNotification	status: Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
4.14	CSMS → CS	ChangeConfiguration (Heartbeat)	status (Accepted)	[CSMS] -> ChangeConfig: Heartbeat=600 [CS] -> Conf: status=Accepted	Pass
4.15	CSMS → CS	ChangeConfiguration (MeterValue)	status (Accepted)	[CSMS] -> ChangeConfig: MeterValue=60 [CS] -> Conf: status=Accepted	Pass
4.16	CSMS → CS	ChangeConfiguration (LocalAuth)	status (Accepted)	[CSMS] -> ChangeConfig: LocalAuth=True [CS] -> Conf: status=Accepted	Pass
4.17	CS → CSMS	StatusNotification	status: Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
4.18	CS → CSMS	StartTransaction	status (Accepted)	[CSMS] <- StartTx [CSMS] -> Conf: Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
4.19	CSMS → CS	Reset (Hard)	status (Accepted)	[CSMS] -> Reset (Hard) [CS] -> Conf: Accepted	Pass
4.20	CS → CSMS	StopTransaction (Reset)	status (Accepted)	[CSMS] <- StopTx: HardReset [CSMS] -> Conf: status=Accepted	Pass
4.21	CS → CSMS	BootNotification	Post-reboot checks	[CSMS] <- BootNotification [CSMS] -> Conf: status=Accepted	Pass
5. Reservation Verification					
5.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
5.2	CSMS → CS	ReserveNow	status (Accepted)	[CSMS] -> ReserveNow [CS] -> Conf: status=Accepted	Pass
5.3	CS → CSMS	StatusNotification	Reserved	[CSMS] <- Status: Reserved [CSMS] -> Conf: {}	Pass
5.4	CSMS → CS	CancelReservation	status (Accepted)	[CSMS] -> CancelReservation [CS] -> Conf: status=Accepted	Pass
5.5	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
5.6	CSMS → CS	ReserveNow (Short)	status (Accepted)	[CSMS] -> ReserveNow [CS] -> Conf: status=Accepted	Pass
5.7	CS → CSMS	StatusNotification	Reserved	[CSMS] <- Status: Reserved [CSMS] -> Conf: {}	Pass
5.8	CS → CSMS	Wait Expiry	(Internal)	Waiting for expiry...	Pass
5.9	CS → CSMS	StatusNotification	Available (Auto)	[CSMS] <- Status: Available	Pass
5.10	CSMS → CS	ReserveNow	status (Accepted)	[CSMS] -> ReserveNow [CS] -> Conf: status=Accepted	Pass
5.11	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
5.12	CSMS → CS	RemoteStartTransaction	status (Accepted)	[CSMS] -> RemoteStart [CS] -> Conf: status=Accepted	Pass
5.13	CS → CSMS	StartTransaction	status (Accepted)	[CSMS] <- StartTx [CSMS] -> Conf: Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
5.14	CS → CSMS	StatusNotification	Charging	[CSMS] <- Status: Charging [CSMS] -> Conf: {}	Pass
5.15	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
5.16	CSMS → CS	RemoteStopTransaction	status (Accepted)	[CSMS] -> RemoteStop [CS] -> Conf: status=Accepted	Pass
5.17	CS → CSMS	StopTransaction	status (Accepted)	[CSMS] <- StopTx [CSMS] -> Conf: Accepted	Pass
5.18	CS → CSMS	StatusNotification	Finishing	[CSMS] <- Status: Finishing [CSMS] -> Conf: {}	Pass
5.19	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
6. Charging Profile Verification					
6.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
6.2	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
6.3	CS → CSMS	Authorize	Accepted	[CSMS] <- Authorize [CSMS] -> Conf: status=Accepted	Pass
6.4	CS → CSMS	StartTransaction	Accepted	[CSMS] <- StartTx [CSMS] -> Conf: status=Accepted	Pass
6.5	CS → CSMS	StatusNotification	Charging	[CSMS] <- Status: Charging [CSMS] -> Conf: {}	Pass
6.6	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
6.7	CSMS → CS	SetChargingProfile	status (Accepted)	[CSMS] -> SetChargingProfile [CS] -> Conf: status=Accepted	Pass
6.8	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
6.9	CSMS → CS	SetChargingProfile	status (Accepted)	[CSMS] -> SetChargingProfile [CS] -> Conf: status=Accepted	Pass
6.10	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
6.11	CS → CSMS	StopTransaction	Accepted	[CSMS] <- StopTx [CSMS] -> Conf: status=Accepted	Pass
6.12	CS → CSMS	StatusNotification	Finishing	[CSMS] <- Status: Finishing [CSMS] -> Conf: {}	Pass
6.13	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
6.14	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
6.15	CSMS → CS	RemoteStartTransaction	status (Accepted)	[CSMS] -> RemoteStart [CS] -> Conf: status=Accepted	Pass
6.16	CS → CSMS	StartTransaction	Accepted	[CSMS] <- StartTx [CSMS] -> Conf: status=Accepted	Pass
6.17	CS → CSMS	StatusNotification	Charging	[CSMS] <- Status: Charging [CSMS] -> Conf: {}	Pass
6.18	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
6.19	CSMS → CS	SetChargingProfile	(Accepted)	[CSMS] -> SetChargingProfile [CS] -> Conf: status=Accepted	Pass
6.20	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
6.21	CSMS → CS	SetChargingProfile	(Accepted)	[CSMS] -> SetChargingProfile [CS] -> Conf: status=Accepted	Pass
6.22	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
6.23	CSMS → CS	RemoteStopTransaction	status (Accepted)	[CSMS] -> RemoteStop [CS] -> Conf: status=Accepted	Pass
6.24	CS → CSMS	StopTransaction	Accepted	[CSMS] <- StopTx [CSMS] -> Conf: status=Accepted	Pass
6.25	CS → CSMS	StatusNotification	Finishing	[CSMS] <- Status: Finishing [CSMS] -> Conf: {}	Pass
6.26	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass

7. Abnormal Operation Verification

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
7.1 Remote Start (Unplugged)					
7.1.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.1.2	CSMS → CS	RemoteStartTransaction	Accepted	[CSMS] -> RemoteStart [CS] -> Conf: status=Accepted	Pass
7.2 Concurrent Remote Start					
7.2.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.2.2	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
7.2.3	CSMS → CS	RemoteStart (Conn 1)	Rejected	[CSMS] -> RemoteStart (1) [CS] -> Conf: status=Rejected	Pass
7.2.4	CSMS → CS	RemoteStart (Conn 2)	Rejected	[CSMS] -> RemoteStart (2) [CS] -> Conf: status=Rejected	Pass
7.2.5	CS → CSMS	StartTransaction	Accepted	[CSMS] <- StartTx [CSMS] -> Conf: Accepted	Pass
7.2.6	CS → CSMS	StatusNotification	Charging	[CSMS] <- Status: Charging [CSMS] -> Conf: {}	Pass
7.2.7	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
7.2.8	CSMS → CS	RemoteStopTransaction	Accepted	[CSMS] -> RemoteStop [CS] -> Conf: status=Accepted	Pass
7.2.9	CS → CSMS	StopTransaction	Accepted	[CSMS] <- StopTx [CSMS] -> Conf: Accepted	Pass
7.2.10	CS → CSMS	StatusNotification	Finishing	[CSMS] <- Status: Finishing [CSMS] -> Conf: {}	Pass
7.2.11	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.3 Swap Card					
7.3.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
7.3.2	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
7.3.3	CS → CSMS	Authorize (Invalid)	Rejected	[CSMS] <- Authorize (Invalid) [CSMS] -> Conf: status=Invalid	Pass
7.3.4	CS → CSMS	Authorize (Valid)	Accepted	[CSMS] <- Authorize (Valid) [CSMS] -> Conf: status=Accepted	Pass
7.3.5	CS → CSMS	StartTransaction	Accepted	[CSMS] <- StartTx [CSMS] -> Conf: Accepted	Pass
7.3.6	CS → CSMS	StatusNotification	Charging	[CSMS] <- Status: Charging [CSMS] -> Conf: {}	Pass
7.3.7	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
7.3.8	CS → CSMS	Authorize (Invalid Stop)	Rejected	[CSMS] <- Authorize (Invalid) [CSMS] -> Conf: Invalid	Pass
7.3.9	CS → CSMS	StopTransaction	Accepted	[CSMS] <- StopTx [CSMS] -> Conf: Accepted	Pass
7.3.10	CS → CSMS	StatusNotification	Finishing	[CSMS] <- Status: Finishing [CSMS] -> Conf: {}	Pass
7.3.11	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.4 Emergency Stop					
7.4.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.4.2	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
7.4.3	CSMS → CS	RemoteStartTransaction	Accepted	[CSMS] -> RemoteStart [CS] -> Conf: status=Accepted	Pass
7.4.4	CS → CSMS	StartTransaction	Accepted	[CSMS] <- StartTx [CSMS] -> Conf: Accepted	Pass
7.4.5	CS → CSMS	StatusNotification	Charging	[CSMS] <- Status: Charging [CSMS] -> Conf: {}	Pass
7.4.6	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
7.4.7	CS → CSMS	StopTransaction	Accepted	[CSMS] <- StopTx (Emergency) [CSMS] -> Conf: Accepted	Pass
7.4.8	CS → CSMS	StatusNotification	Faulted	[CSMS] <- Status: Faulted [CSMS] -> Conf: {}	Pass
7.4.9	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.4.10	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.5 Open Door					
7.5.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.5.2	CS → CSMS	StatusNotification	Faulted	[CSMS] <- Status: Faulted (Door) [CSMS] -> Conf: {}	Pass
7.5.3	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.6 Power Loss (Single)					
7.6.1	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.6.2	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
7.6.3	CSMS → CS	RemoteStartTransaction	Accepted	[CSMS] -> RemoteStart [CS] -> Conf: status=Accepted	Pass
7.6.4	CS → CSMS	StartTransaction	Accepted	[CSMS] <- StartTx [CSMS] -> Conf: Accepted	Pass
7.6.5	CS → CSMS	StatusNotification	Charging	[CSMS] <- Status: Charging [CSMS] -> Conf: {}	Pass
7.6.6	CS → CSMS	MeterValues	Received	[CSMS] <- MeterValues [CSMS] -> Conf: {}	Pass
7.6.7	CS → CSMS	StopTransaction	Accepted	[CSMS] <- StopTx (PowerLoss) [CSMS] -> Conf: Accepted	Pass
7.6.8	CS → CSMS	BootNotification	Accepted	[CSMS] <- BootNotification [CSMS] -> Conf: Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
7.6.9	CS → CSMS	StatusNotification	Preparing	[CSMS] <- Status: Preparing [CSMS] -> Conf: {}	Pass
7.6.10	CS → CSMS	StatusNotification	Available	[CSMS] <- Status: Available [CSMS] -> Conf: {}	Pass
7.7 Local List (Offline)					
7.7.1	N/A	Disconnect LAN	Simulated	Simulated Disconnection	Pass
7.7.2	CS → CSMS	StartTransaction	Accepted	[CSMS] <- StartTx (Offline) [CSMS] -> Conf: Accepted	Pass
7.7.3	CS → CSMS	StopTransaction	Accepted	[CSMS] <- StopTx (Local) [CSMS] -> Conf: Accepted	Pass
8. Dual Connector Verification					
8.1 Concurrent Remote Start (Dual)					
8.1.1	CS → CSMS	StatusNotification (Un-plug)	Available	[CSMS] <- Status: Available	Pass
8.1.2	CS → CSMS	StatusNotification (Plug Both)	Preparing	[CSMS] <- Status: Preparing (1&2)	Pass
8.1.3	CSMS → CS	RemoteStart (Conn 1)	Accepted	[CSMS] -> RemoteStart (1) [CS] -> Conf: Accepted	Pass
8.1.4	CSMS → CS	RemoteStart (Conn 2)	Accepted	[CSMS] -> RemoteStart (2) [CS] -> Conf: Accepted	Pass
8.1.5	CS → CSMS	StartTransaction (Conn 1)	Accepted	[CSMS] <- StartTx (1) [CSMS] -> Conf: Accepted	Pass
8.1.6	CS → CSMS	StatusNotification (Conn 1)	Charging	[CSMS] <- Status: Charging	Pass
8.1.7	CS → CSMS	StartTransaction (Conn 2)	Accepted	[CSMS] <- StartTx (2) [CSMS] -> Conf: Accepted	Pass
8.1.8	CS → CSMS	StatusNotification (Conn 2)	Charging	[CSMS] <- Status: Charging	Pass
8.1.9	CS → CSMS	MeterValues (Conn 1)	Received	[CSMS] <- MeterValues (1)	Pass
8.1.10	CS → CSMS	MeterValues (Conn 2)	Received	[CSMS] <- MeterValues (2)	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
8.1.11	CSMS → CS	RemoteStop (Conn 1)	Accepted	[CSMS] -> RemoteStop (1) [CS] -> Conf: Accepted	Pass
8.1.12	CS → CSMS	StopTransaction (Conn 1)	Accepted	[CSMS] <- StopTx (1) [CSMS] -> Conf: Accepted	Pass
8.1.13	CS → CSMS	StatusNotification (Conn 1)	Finishing	[CSMS] <- Status: Finishing	Pass
8.1.14	CS → CSMS	StatusNotification (Conn 1)	Available	[CSMS] <- Status: Available	Pass
8.1.15	CSMS → CS	RemoteStop (Conn 2)	Accepted	[CSMS] -> RemoteStop (2) [CS] -> Conf: Accepted	Pass
8.1.16	CS → CSMS	StopTransaction (Conn 2)	Accepted	[CSMS] <- StopTx (2) [CSMS] -> Conf: Accepted	Pass
8.1.17	CS → CSMS	StatusNotification (Conn 2)	Finishing	[CSMS] <- Status: Finishing	Pass
8.1.18	CS → CSMS	StatusNotification (Conn 2)	Available	[CSMS] <- Status: Available	Pass
8.2 Shared Emergency Stop (Dual)					
8.2.1	CS → CSMS	StatusNotification (Unplug)	Available	[CSMS] <- Status: Available (Both)	Pass
8.2.2	CS → CSMS	StatusNotification (Conn 1)	Preparing	[CSMS] <- Status: Preparing (1)	Pass
8.2.3	CSMS → CS	RemoteStart (Conn 1)	Accepted	[CSMS] -> RemoteStart (1) [CS] -> Conf: Accepted	Pass
8.2.4	CS → CSMS	StartTransaction (Conn 1)	Accepted	[CSMS] <- StartTx (1) [CSMS] -> Conf: Accepted	Pass
8.2.5	CS → CSMS	StatusNotification (Conn 1)	Charging	[CSMS] <- Status: Charging	Pass
8.2.6	CS → CSMS	MeterValues (Conn 1)	Received	[CSMS] <- MeterValues (1)	Pass
8.2.7	CS → CSMS	StatusNotification (Conn 2)	Preparing	[CSMS] <- Status: Preparing (2)	Pass
8.2.8	CSMS → CS	RemoteStart (Conn 2)	Accepted	[CSMS] -> RemoteStart (2) [CS] -> Conf: Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
8.2.9	CS → CSMS	StartTransaction (Conn 2)	Accepted	[CSMS] <- StartTx (2) [CSMS] -> Conf: Accepted	Pass
8.2.10	CS → CSMS	StatusNotification (Conn 2)	Charging	[CSMS] <- Status: Charging	Pass
8.2.11	CS → CSMS	MeterValues (Conn 2)	Received	[CSMS] <- MeterValues (2)	Pass
8.2.12	CS → CSMS	StopTransaction (Conn 1)	Accepted	[CSMS] <- StopTx (1) (Emergency) [CSMS] -> Conf: Accepted	Pass
8.2.13	CS → CSMS	StopTransaction (Conn 2)	Accepted	[CSMS] <- StopTx (2) (Emergency) [CSMS] -> Conf: Accepted	Pass
8.2.14	CS → CSMS	StatusNotification (Both)	Faulted	[CSMS] <- Status: Faulted (Info: Emergency)	Pass
8.2.15	CS → CSMS	StatusNotification (Unplug)	Faulted	[CSMS] <- Status: Faulted (Persist)	Pass
8.2.16	CS → CSMS	StatusNotification (Release)	Available	[CSMS] <- Status: Available	Pass
8.3 Power Loss (Dual)					
8.3.1	CS → CSMS	StatusNotification (Unplug)	Available	[CSMS] <- Status: Available	Pass
8.3.2	CS → CSMS	StatusNotification (Conn 1)	Preparing	[CSMS] <- Status: Preparing	Pass
8.3.3	CSMS → CS	RemoteStart (Conn 1)	Accepted	[CSMS] -> RemoteStart (1) [CS] -> Conf: Accepted	Pass
8.3.4	CS → CSMS	StartTransaction (Conn 1)	Accepted	[CSMS] <- StartTx (1) [CSMS] -> Conf: Accepted	Pass
8.3.5	CS → CSMS	StatusNotification (Conn 1)	Charging	[CSMS] <- Status: Charging	Pass
8.3.6	CS → CSMS	MeterValues (Conn 1)	Received	[CSMS] <- MeterValues (1)	Pass
8.3.7	CS → CSMS	StatusNotification (Conn 2)	Preparing	[CSMS] <- Status: Preparing	Pass
8.3.8	CSMS → CS	RemoteStart (Conn 2)	Accepted	[CSMS] -> RemoteStart (2) [CS] -> Conf: Accepted	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
8.3.9	CS → CSMS	StartTransaction (Conn 2)	Accepted	[CSMS] <- StartTx (2) [CSMS] -> Conf: Accepted	Pass
8.3.10	CS → CSMS	StatusNotification (Conn 2)	Charging	[CSMS] <- Status: Charging	Pass
8.3.11	CS → CSMS	MeterValues (Conn 2)	Received	[CSMS] <- MeterValues (2)	Pass
8.3.12	CS → CSMS	StopTransaction (Conn 1)	Accepted	[CSMS] <- StopTx (1) (PowerLoss) [CSMS] -> Conf: Accepted	Pass
8.3.13	CS → CSMS	StopTransaction (Conn 2)	Accepted	[CSMS] <- StopTx (2) (PowerLoss) [CSMS] -> Conf: Accepted	Pass
8.3.14	CS → CSMS	BootNotification	Accepted	[CSMS] <- BootNotification [CSMS] -> Conf: Accepted	Pass
8.3.15	CS → CSMS	StatusNotification (Plug Both)	Preparing	N/A (Skipped in Test)	Skipped
8.3.16	CS → CSMS	StatusNotification (Unplug Both)	Available	[CSMS] <- Status(1): Avail [CSMS] -> Conf: {} [CSMS] <- Status(2): Avail [CSMS] -> Conf: {}	Pass
9. MEA Specific Configuration					
9.1	CSMS → CS	ChangeConfiguration (V2G)	status (Accepted)	[CSMS] -> ChangeConfig: MEA.V2G=True [CS] -> Conf: status=Accepted	Pass
9.2	CSMS → CS	ChangeConfig (Meter-Sample)	status (Accepted)	[CSMS] -> ChangeConfig: Sample=60 [CS] -> Conf: status=Accepted	Pass
9.3	CSMS → CS	ChangeConfiguration (PowerDemand 2k)	status (Accepted)	[CSMS] -> ChangeConfig: PowerDemand=2000 [CS] -> Conf: status=Accepted	Pass
9.4	CSMS → CS	ChangeConfiguration (PowerDemand -5k)	status (Accepted)	[CSMS] -> ChangeConfig: PowerDemand=-5000 [CS] -> Conf: status=Accepted	Pass
9.5	CSMS → CS	ChangeConfiguration (PowerDemand 0)	status (Accepted)	[CSMS] -> ChangeConfig: PowerDemand=0 [CS] -> Conf: status=Accepted	Pass
10. Summary Verification					
10.1	CSMS	Log Verification	All logs present	See Appendix A	Pass
10.19	CSMS → CS	GetDiagnostics	fileName Returned	[CSMS] -> GetDiagnostics [CS] -> Conf: fileName=...	Pass

Item	Direction	Message	Requirement / Include	Log Message / Proof	Result
10.20	CSMS → CS	UpdateFirmware	Request Sent	[CSMS] -> UpdateFirmware [CS] -> Conf: {}	Pass
10.21	CSMS → CS	SendLocalList	Accepted	[CSMS] -> SendLocalList [CS] -> Conf: status=Accepted	Pass
10.22	CSMS → CS	GetLocalListVersion	Version 1	[CSMS] -> GetLocalListVersion [CS] -> Conf: listVersion=1	Pass
11. Other Verification					
11.1	CS → CSMS	LAN Disconnect	Reconnect	[CSMS] <- BootNotification (Reconnect)	Pass

A Raw Test Logs

```
INFO:websockets.server:server listening on [::1]:9000
INFO:websockets.server:server listening on 127.0.0.1:9000
INFO:MockSystemTest:Mock CSMS started on localhost:9000
INFO:websockets.server:connection open
INFO:MockSystemTest:CSMS: Client connected! CP_TEST
INFO:MockSystemTest:
--- 1. Configuration Verification ---
INFO:ocpp:CP_TEST: send [2,"6b90c18d-5cba-4150-be00-86c9aba90e4a","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2,"6b90c18d-5cba-4150-be00-86c9aba90e4a","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:42: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to
    represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    current_time=datetime.utcnow().isoformat(),
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:26:41.510476', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3,"6b90c18d-5cba-4150-be00-86c9aba90e4a",{"current_time":"2026-01-05T15:26:41.510476","interval":10,"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"6b90c18d-5cba-4150-be00-86c9aba90e4a",{"current_time":"2026-01-05T15:26:41.510476","interval":10,"status":"Accepted"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: send [2,"471f5b2a-f4ac-424f-a5f1-44a79636c148","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"471f5b2a-f4ac-424f-a5f1-44a79636c148","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"471f5b2a-f4ac-424f-a5f1-44a79636c148",{}]
INFO:ocpp:CP_TEST: receive message [3,"471f5b2a-f4ac-424f-a5f1-44a79636c148",{}]
INFO:ocpp:CP_TEST: send [2,"167c56bd-2edc-4d1c-b461-98b9533538f0","TriggerMessage",{"requestedMessage":"BootNotification"}]
INFO:ocpp:CP_TEST: receive message [2,"167c56bd-2edc-4d1c-b461-98b9533538f0","TriggerMessage",{"requestedMessage":"BootNotification"}]
INFO:ocpp_16_interface:Received TriggerMessage for BootNotification
INFO:ocpp:CP_TEST: send [2,"d9e77db4-bdab-4851-9847-136e26fcb86c","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2,"d9e77db4-bdab-4851-9847-136e26fcb86c","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:ocpp:CP_TEST: send [3,"167c56bd-2edc-4d1c-b461-98b9533538f0",{"status":"Accepted"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:26:41.515950', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3,"d9e77db4-bdab-4851-9847-136e26fcb86c",{"current_time":"2026-01-05T15:26:41.515950","interval":10,"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"167c56bd-2edc-4d1c-b461-98b9533538f0",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"d9e77db4-bdab-4851-9847-136e26fcb86c",{"current_time":"2026-01-05T15:26:41.515950","interval":10,"status":"Accepted"}]
INFO:MockSystemTest:1.3 Trigger BootNotification: Accepted
INFO:ocpp:CP_TEST: send [2,"d9f7dc52-d89b-4e74-8b82-2a6604177c7b","TriggerMessage",{"requestedMessage":"StatusNotification"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2,"d9f7dc52-d89b-4e74-8b82-2a6604177c7b","TriggerMessage",{"requestedMessage":"StatusNotification"}]
INFO:ocpp_16_interface:Received TriggerMessage for StatusNotification
INFO:ocpp:CP_TEST: send [3,"d9f7dc52-d89b-4e74-8b82-2a6604177c7b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"e755ad07-0431-4e7e-872c-8a0bc18a6c1e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [3,"d9f7dc52-d89b-4e74-8b82-2a6604177c7b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"e755ad07-0431-4e7e-872c-8a0bc18a6c1e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:1.5 Trigger Status: Accepted
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [2,"d9550b41-a944-4fb5-aa01-ec91c6bf791","TriggerMessage",{"requestedMessage":"MeterValues"}]
INFO:ocpp:CP_TEST: send [3,"e755ad07-0431-4e7e-872c-8a0bc18a6c1e",{}]
INFO:ocpp:CP_TEST: receive message [2,"d9550b41-a944-4fb5-aa01-ec91c6bf791","TriggerMessage",{"requestedMessage":"MeterValues"}]
INFO:ocpp_16_interface:Received TriggerMessage for MeterValues
ERROR:asyncio:Task exception was never retrieved
future: <Task finished name='Task-10' coro=<Ocpp16Interface.send_meter_values_one_off() done, defined at /Users/rus/development/MEA-V2G/Ocpp16Interface.py:294> exception=AttributeError("'Ocpp16Interface' object
    has no attribute 'transaction_id'")>
Traceback (most recent call last):
  File "/Users/rus/development/MEA-V2G/Ocpp16Interface.py", line 308, in send_meter_values_one_off
```

```
transaction_id=self.transaction_id,
.....
AttributeError: 'Ocpp16Interface' object has no attribute 'transaction_id'. Did you mean: 'transactions'?
INFO:ocpp:CP_TEST: send [3,"d9550b41-a944-4fb5-aa01-ec91c6bf791",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"e755ad07-0431-4e7e-872c-8a0bc18a6c1e",{}]
INFO:ocpp:CP_TEST: receive message [3,"d9550b41-a944-4fb5-aa01-ec91c6bf791",{"status":"Accepted"}]
INFO:MockSystemTest:1.7 Trigger MeterValues: Accepted
INFO:ocpp:CP_TEST: send [2,"90af7aea-6ed2-4f9e-82b1-ac9c56afaa15",{"key":"HeartbeatInterval","value":"600"}]
INFO:ocpp:CP_TEST: receive message [2,"90af7aea-6ed2-4f9e-82b1-ac9c56afaa15",{"key":"HeartbeatInterval","value":"600"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for HeartbeatInterval to 600
INFO:ocpp:CP_TEST: send [3,"90af7aea-6ed2-4f9e-82b1-ac9c56afaa15",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"90af7aea-6ed2-4f9e-82b1-ac9c56afaa15",{"status":"Accepted"}]
INFO:MockSystemTest:1.10 Change Heartbeat: Accepted
INFO:ocpp:CP_TEST: send [2,"a167f463-1383-4c29-a2fc-f8eb061e9c21",{"key":["HeartbeatInterval"]}]]
INFO:ocpp:CP_TEST: receive message [2,"a167f463-1383-4c29-a2fc-f8eb061e9c21",{"key":["HeartbeatInterval"]}]]
INFO:ocpp_16_interface:Received GetConfiguration for None
INFO:ocpp:CP_TEST: send [3,"a167f463-1383-4c29-a2fc-f8eb061e9c21",{"configurationKey":[{"key":"HeartbeatInterval","readonly":false,"value":"600"},{"key":"ConnectionTimeOut","readonly":false,"value":"60"},{"key":"MeterValueSampleInterval","readonly":false,"value":"60"},{"key":"LocalAuthorizeOffline","readonly":false,"value":"False"},{"key":"UnlockConnectorOnEVSideDisconnect","readonly":false,"value":"False"},{"key":"AutoCharge","readonly":false,"value":"False"},{"key":"MEA_V2G_PowerDemand","readonly":false,"value":"0"}]]]
INFO:ocpp:CP_TEST: receive message [3,"a167f463-1383-4c29-a2fc-f8eb061e9c21",{"configurationKey":[{"key":"HeartbeatInterval","readonly":false,"value":"600"},{"key":"ConnectionTimeOut","readonly":false,"value":"60"},{"key":"MeterValueSampleInterval","readonly":false,"value":"60"},{"key":"LocalAuthorizeOffline","readonly":false,"value":"False"},{"key":"UnlockConnectorOnEVSideDisconnect","readonly":false,"value":"False"},{"key":"AutoCharge","readonly":false,"value":"False"},{"key":"MEA_V2G_PowerDemand","readonly":false,"value":"0"}]]]
INFO:MockSystemTest:PASS: HeartbeatInterval verified.
INFO:ocpp:CP_TEST: send [2,"9e1120ae-66d4-4ca5-9644-f392247d20f2",{"key":"MeterValueSampleInterval","value":"2"}]
INFO:ocpp:CP_TEST: receive message [2,"9e1120ae-66d4-4ca5-9644-f392247d20f2",{"key":"MeterValueSampleInterval","value":"2"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for MeterValueSampleInterval to 2
INFO:ocpp:CP_TEST: send [3,"9e1120ae-66d4-4ca5-9644-f392247d20f2",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"9e1120ae-66d4-4ca5-9644-f392247d20f2",{"status":"Accepted"}]
INFO:MockSystemTest:1.11 Change MeterValue: Accepted
INFO:ocpp:CP_TEST: send [2,"f9028312-3bb1-4337-bc46-d7323f40afff",{"connectorId":1}]
INFO:ocpp:CP_TEST: receive message [2,"f9028312-3bb1-4337-bc46-d7323f40afff",{"connectorId":1}]
INFO:ocpp_16_interface:Received UnlockConnector for 1
INFO:ocpp:CP_TEST: send [3,"f9028312-3bb1-4337-bc46-d7323f40afff",{"status":"Unlocked"}]
INFO:ocpp:CP_TEST: receive message [3,"f9028312-3bb1-4337-bc46-d7323f40afff",{"status":"Unlocked"}]
INFO:MockSystemTest:1.12 UnlockConnector: Unlocked
INFO:ocpp:CP_TEST: send [2,"c2ef3d35-216f-428a-bebd-dbccfc45effe",{"connectorId":1,"type":"Inoperative"}]
INFO:ocpp:CP_TEST: receive message [2,"c2ef3d35-216f-428a-bebd-dbccfc45effe",{"connectorId":1,"type":"Inoperative"}]
INFO:ocpp_16_interface:Received ChangeAvailability Inoperative for 1
INFO:ocpp:CP_TEST: send [3,"c2ef3d35-216f-428a-bebd-dbccfc45effe",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"c2ef3d35-216f-428a-bebd-dbccfc45effe",{"status":"Accepted"}]
INFO:MockSystemTest:1.13 ChangeAvailability Inoperative: Accepted
INFO:ocpp:CP_TEST: send [2,"1443782d-b9f5-4b7f-b78a-e008ec66dab7",{"connectorId":1,"type":"Operative"}]
INFO:ocpp:CP_TEST: receive message [2,"1443782d-b9f5-4b7f-b78a-e008ec66dab7",{"connectorId":1,"type":"Operative"}]
INFO:ocpp_16_interface:Received ChangeAvailability Operative for 1
INFO:ocpp:CP_TEST: send [3,"1443782d-b9f5-4b7f-b78a-e008ec66dab7",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"1443782d-b9f5-4b7f-b78a-e008ec66dab7",{"status":"Accepted"}]
INFO:MockSystemTest:1.15 ChangeAvailability Operative: Accepted
INFO:ocpp:CP_TEST: send [2,"ecae36de-a4f5-4c0c-9a2b-594dae91c499",{"location":"ftp://example.com"}]
INFO:ocpp:CP_TEST: receive message [2,"ecae36de-a4f5-4c0c-9a2b-594dae91c499",{"location":"ftp://example.com"}]
INFO:ocpp_16_interface:Received GetDiagnostics for ftp://example.com
INFO:ocpp:CP_TEST: send [3,"ecae36de-a4f5-4c0c-9a2b-594dae91c499",{"fileName":"diagnostics.log"}]
INFO:ocpp:CP_TEST: receive message [3,"ecae36de-a4f5-4c0c-9a2b-594dae91c499",{"fileName":"diagnostics.log"}]
INFO:MockSystemTest:1.17 GetDiagnostics: diagnostics.log
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:252: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    res = await connected_csms.call(call.UpdateFirmware(location="ftp://firmware.bin", retrieve_date=datetime.utcnow().isoformat()))
INFO:ocpp:CP_TEST: send [2,"60c5f697-48e9-494f-aff3-baf366a1a528",{"location":"ftp://firmware.bin","retrieveDate":"2026-01-05T15:26:41.535484"}]
INFO:ocpp:CP_TEST: receive message [2,"60c5f697-48e9-494f-aff3-baf366a1a528",{"location":"ftp://firmware.bin","retrieveDate":"2026-01-05T15:26:41.535484"}]
INFO:ocpp_16_interface:Received UpdateFirmware from ftp://firmware.bin
INFO:ocpp:CP_TEST: send [3,"60c5f697-48e9-494f-aff3-baf366a1a528",{}]
```

```
INFO:ocpp:CP_TEST: receive message [3,"60c5f697-48e9-494f-aff3-baf366a1a528",{}]]
INFO:MockSystemTest:1.19 UpdateFirmware: UpdateFirmware()
INFO:ocpp:CP_TEST: send [2,"c04baaa8-864d-4d88-bc3c-aa4a0cad5873","ChangeConfiguration",{"key":"LocalAuthorizeOffline","value":"True"}]
INFO:ocpp:CP_TEST: receive message [2,"c04baaa8-864d-4d88-bc3c-aa4a0cad5873","ChangeConfiguration",{"key":"LocalAuthorizeOffline","value":"True"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for LocalAuthorizeOffline to True
INFO:ocpp:CP_TEST: send [3,"c04baaa8-864d-4d88-bc3c-aa4a0cad5873",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"c04baaa8-864d-4d88-bc3c-aa4a0cad5873",{"status":"Accepted"}]
INFO:MockSystemTest:1.21 LocalAuthOffline: Accepted
INFO:ocpp:CP_TEST: send [2,"09615cda-264d-4819-b731-2f2e3aa81ecb","SendLocalList",{"listVersion":1,"updateType":"Full","localAuthorizationList":[]}]
INFO:ocpp:CP_TEST: receive message [2,"09615cda-264d-4819-b731-2f2e3aa81ecb","SendLocalList",{"listVersion":1,"updateType":"Full","localAuthorizationList":[]}]
INFO:ocpp_16_interface:Received SendLocalList
INFO:ocpp:CP_TEST: send [3,"09615cda-264d-4819-b731-2f2e3aa81ecb",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"09615cda-264d-4819-b731-2f2e3aa81ecb",{"status":"Accepted"}]
INFO:MockSystemTest:1.22 SendLocalList: Accepted
INFO:ocpp:CP_TEST: send [2,"c8e21f56-f626-43b1-af1d-ec4d5a111d36","GetLocalListVersion",{}]]
INFO:ocpp:CP_TEST: receive message [2,"c8e21f56-f626-43b1-af1d-ec4d5a111d36","GetLocalListVersion",{}]]
INFO:ocpp_16_interface:Received GetLocalListVersion
INFO:ocpp:CP_TEST: send [3,"c8e21f56-f626-43b1-af1d-ec4d5a111d36",{"listVersion":1}]
INFO:ocpp:CP_TEST: receive message [3,"c8e21f56-f626-43b1-af1d-ec4d5a111d36",{"listVersion":1}]
INFO:MockSystemTest:1.23 GetLocalListVersion: 1
INFO:ocpp:CP_TEST: send [2,"ab8d438b-8c19-4fdc-a4bc-a8c5063658b2","ClearCache",{}]]
INFO:ocpp:CP_TEST: receive message [2,"ab8d438b-8c19-4fdc-a4bc-a8c5063658b2","ClearCache",{}]]
INFO:ocpp_16_interface:Received ClearCache
INFO:ocpp:CP_TEST: send [3,"ab8d438b-8c19-4fdc-a4bc-a8c5063658b2",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"ab8d438b-8c19-4fdc-a4bc-a8c5063658b2",{"status":"Accepted"}]
INFO:MockSystemTest:1.24 ClearCache: Accepted
INFO:MockSystemTest:
--- 2. Auto Charge Verification ---
INFO:ocpp:CP_TEST: send [2,"520b344a-d788-46c6-a738-1146ab3e2474","ChangeConfiguration",{"key":"AutoCharge","value":"False"}]
INFO:ocpp:CP_TEST: receive message [2,"520b344a-d788-46c6-a738-1146ab3e2474","ChangeConfiguration",{"key":"AutoCharge","value":"False"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for AutoCharge to False
INFO:ocpp:CP_TEST: send [3,"520b344a-d788-46c6-a738-1146ab3e2474",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"520b344a-d788-46c6-a738-1146ab3e2474",{"status":"Accepted"}]
INFO:MockSystemTest:2.1 Disable AutoCharge: Accepted
INFO:ocpp:CP_TEST: send [2,"ba765b16-a62f-4cc9-a175-6fd2f9be1301","ChangeConfiguration",{"key":"AutoCharge","value":"True"}]
INFO:ocpp:CP_TEST: receive message [2,"ba765b16-a62f-4cc9-a175-6fd2f9be1301","ChangeConfiguration",{"key":"AutoCharge","value":"True"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for AutoCharge to True
INFO:ocpp:CP_TEST: send [3,"ba765b16-a62f-4cc9-a175-6fd2f9be1301",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"ba765b16-a62f-4cc9-a175-6fd2f9be1301",{"status":"Accepted"}]
INFO:MockSystemTest:2.4 Enable AutoCharge: Accepted
INFO:ocpp:CP_TEST: send [2,"7181bcba-bffa-4cac-94c3-47cfae681c90","RemoteStopTransaction",{"transactionId":999}]
INFO:ocpp:CP_TEST: receive message [2,"7181bcba-bffa-4cac-94c3-47cfae681c90","RemoteStopTransaction",{"transactionId":999}]
INFO:ocpp_16_interface:Received RemoteStopTransaction for transaction_id: 999
INFO:ocpp:CP_TEST: send [3,"7181bcba-bffa-4cac-94c3-47cfae681c90",{"status":"Rejected"}]
INFO:ocpp:CP_TEST: receive message [3,"7181bcba-bffa-4cac-94c3-47cfae681c90",{"status":"Rejected"}]
INFO:MockSystemTest:2.10 RemoteStopTransaction: Rejected
INFO:MockSystemTest:--- 2.x Extended Auto Charge Verification ---
INFO:ocpp:CP_TEST: send [2,"d4661aec-4279-4d35-a571-93c3b992732a","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"d4661aec-4279-4d35-a571-93c3b992732a","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"d4661aec-4279-4d35-a571-93c3b992732a",{}]]
INFO:ocpp:CP_TEST: receive message [3,"d4661aec-4279-4d35-a571-93c3b992732a",{}]]
INFO:MockSystemTest:2.14 StatusNotification (Plug): Sent Preparing
INFO:ocpp:CP_TEST: send [2,"e9b90ea3-c86b-4b6e-9bbf-e97e28a613da","Authorize",{"idTag":"VID:12345678"}]
INFO:ocpp:CP_TEST: receive message [2,"e9b90ea3-c86b-4b6e-9bbf-e97e28a613da","Authorize",{"idTag":"VID:12345678"}]
ERROR:ocpp:Error while handling request '<Call - unique_id=e9b90ea3-c86b-4b6e-9bbf-e97e28a613da, action=Authorize, payload={'idTag': 'VID:12345678'}>'
Traceback (most recent call last):
  File "/Users/rus/Library/Python/3.12/lib/python/site-packages/ocpp/charge_point.py", line 294, in _handle_call
    handlers = self.route_map[msg.action]
```

```
~~~~~  
KeyError: 'Authorize'
```

During handling of the above exception, another exception occurred:

Traceback (most recent call last):

```
File "/Users/rus/Library/Python/3.12/lib/python/site-packages/ocpp/charge_point.py", line 272, in route_message
    await self._handle_call(msg)
File "/Users/rus/Library/Python/3.12/lib/python/site-packages/ocpp/charge_point.py", line 296, in _handle_call
    _raise_key_error(msg.action, self._ocpp_version)
File "/Users/rus/Library/Python/3.12/lib/python/site-packages/ocpp/charge_point.py", line 175, in _raise_key_error
    raise NotImplementedError(
ocpp.exceptions.NotImplementedError: NotImplementedError: Request Action is recognized but not supported by the receiver, {'cause': 'No handler for Authorize registered.'}
INFO:ocpp:CP_TEST: send [4,"e9b90ea3-c86b-4b6e-9bbf-e97e28a613da","NotImplemented","Request Action is recognized but not supported by the receiver",{"cause":"No handler for Authorize registered."}]
INFO:ocpp:CP_TEST: receive message [4,"e9b90ea3-c86b-4b6e-9bbf-e97e28a613da","NotImplemented","Request Action is recognized but not supported by the receiver",{"cause":"No handler for Authorize registered."}]
WARNING:ocpp:Received a CALLError: <CallError - unique_id=e9b90ea3-c86b-4b6e-9bbf-e97e28a613da, error_code=NotImplemented, error_description=Request Action is recognized but not supported by the receiver,
    error_details={'cause': 'No handler for Authorize registered.'}>
INFO:ocpp:CP_TEST: send [2,"4e3e9469-635d-47ce-bd8a-e81ab379170c","Authorize",{"idTag":"VID:12345678"}]
INFO:ocpp:CP_TEST: receive message [2,"4e3e9469-635d-47ce-bd8a-e81ab379170c","Authorize",{"idTag":"VID:12345678"}]
INFO:MockSystemTest:[CSMS] <- Authorize: {'id_tag': 'VID:12345678'}
INFO:MockSystemTest:[CSMS] -> AuthorizeConf: Authorize(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"4e3e9469-635d-47ce-bd8a-e81ab379170c",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"4e3e9469-635d-47ce-bd8a-e81ab379170c",{"idTagInfo":{"status":"Accepted"}}]
INFO:MockSystemTest:2.15 Authorize (Auto Charge): Accepted
INFO:ocpp:CP_TEST: send [2,"27b7e1e8-89cc-41ab-9bb5-10d8461733e3","StartTransaction",{"connectorId":1,"idTag":"VID:12345678","meterStart":0,"timestamp":"2026-01-05T15:26:41.665144"}]
INFO:ocpp:CP_TEST: receive message [2,"27b7e1e8-89cc-41ab-9bb5-10d8461733e3","StartTransaction",{"connectorId":1,"idTag":"VID:12345678","meterStart":0,"timestamp":"2026-01-05T15:26:41.665144"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'VID:12345678', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:41.665144'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"27b7e1e8-89cc-41ab-9bb5-10d8461733e3",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"27b7e1e8-89cc-41ab-9bb5-10d8461733e3",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:2.16 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"7f373557-03da-4a92-a712-195c66295ae0","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"7f373557-03da-4a92-a712-195c66295ae0","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"7f373557-03da-4a92-a712-195c66295ae0",{}]
INFO:ocpp:CP_TEST: receive message [3,"7f373557-03da-4a92-a712-195c66295ae0",{}]
INFO:MockSystemTest:2.17 StatusNotification: Charging
INFO:MockSystemTest:2.18 Waiting for MeterValues...
INFO:ocpp:CP_TEST: send [2,"635ee486-53af-4b84-978d-8491b4ea631f","MeterValues",{"connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:43.667913","sampledValue":[{"value":"230.0","context":"Sample.
Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.Periodic",
format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],{"transactionId":123}}]
INFO:ocpp:CP_TEST: receive message [2,"635ee486-53af-4b84-978d-8491b4ea631f","MeterValues",{"connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:43.667913","sampledValue":[{"value":"230.0","context":"
Sample.Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.
Periodic","format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],{"transactionId":123}}]
INFO:MockSystemTest:[CSMS] <- MeterValues: {'connector_id': 1, 'meter_value': [{'timestamp': '2026-01-05T15:26:43.667913', 'sampled_value': [{'value': '230.0', 'context': 'Sample.Periodic', 'format': 'Raw', '
measurand': 'Voltage', 'unit': 'V'}, {'value': '16.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Current.Import', 'unit': 'A'}, {'value': '3680.0', 'context': 'Sample.Periodic', 'format':
'Raw', 'measurand': 'Power.Active.Import', 'unit': 'W'}]}], 'transaction_id': 123}
INFO:MockSystemTest:[CSMS] -> MeterValuesConf: {}
INFO:ocpp:CP_TEST: send [3,"635ee486-53af-4b84-978d-8491b4ea631f",{}]
INFO:ocpp:CP_TEST: receive message [3,"635ee486-53af-4b84-978d-8491b4ea631f",{}]
INFO:MockSystemTest:PASS: Received 1 MeterValues (2nd Cycle).
INFO:ocpp:CP_TEST: send [2,"f6faf493-4373-4a07-9333-01bd7fc76e70","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"SuspendedEV"}]
INFO:ocpp:CP_TEST: receive message [2,"f6faf493-4373-4a07-9333-01bd7fc76e70","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"SuspendedEV"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'SuspendedEV'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"f6faf493-4373-4a07-9333-01bd7fc76e70",{}]
INFO:ocpp:CP_TEST: receive message [3,"f6faf493-4373-4a07-9333-01bd7fc76e70",{}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'SuspendedEV'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"f6faf493-4373-4a07-9333-01bd7fc76e70",{}]
INFO:ocpp:CP_TEST: receive message [3,"f6faf493-4373-4a07-9333-01bd7fc76e70",{}]
```

```
INFO:MockSystemTest:2.19 StatusNotification: SuspendedEV
INFO:ocpp:CP_TEST: send [2,"25b63b9d-327d-41e5-9c43-212d99efaec4","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:26:44.672434","transactionId":123,"reason":"EVDDisconnected"}}
INFO:ocpp:CP_TEST: receive message [2,"25b63b9d-327d-41e5-9c43-212d99efaec4","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:26:44.672434","transactionId":123,"reason":"EVDDisconnected"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:26:44.672434', 'transaction_id': 123, 'reason': 'EVDDisconnected'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"25b63b9d-327d-41e5-9c43-212d99efaec4",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"25b63b9d-327d-41e5-9c43-212d99efaec4",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:2.20 StopTransaction (Unplug): Sent
INFO:ocpp:CP_TEST: send [2,"d631bdf0-0c58-4400-99c8-deca1806ac07","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"d631bdf0-0c58-4400-99c8-deca1806ac07","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"d631bdf0-0c58-4400-99c8-deca1806ac07",{}]
INFO:ocpp:CP_TEST: receive message [3,"d631bdf0-0c58-4400-99c8-deca1806ac07",{}]
INFO:MockSystemTest:2.21 StatusNotification: Available
INFO:MockSystemTest:
--- 3. Normal Operation Verification (Full Sequence) ---
INFO:ocpp:CP_TEST: send [2,"b0b4e99f-2cae-4886-99e4-be56d0566bf3","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"b0b4e99f-2cae-4886-99e4-be56d0566bf3","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"b0b4e99f-2cae-4886-99e4-be56d0566bf3",{}]
INFO:ocpp:CP_TEST: receive message [3,"b0b4e99f-2cae-4886-99e4-be56d0566bf3",{}]
INFO:MockSystemTest:3.1 Status (Unplug): Available
INFO:ocpp:CP_TEST: send [2,"f4071593-ab05-48f5-94ea-fa902cf2f9d3","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"f4071593-ab05-48f5-94ea-fa902cf2f9d3","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"f4071593-ab05-48f5-94ea-fa902cf2f9d3",{}]
INFO:ocpp:CP_TEST: receive message [3,"f4071593-ab05-48f5-94ea-fa902cf2f9d3",{}]
INFO:MockSystemTest:3.2 Status: Available
INFO:ocpp:CP_TEST: send [2,"24cf4bc7-28b7-45a3-8070-9e87f3517dbb","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"24cf4bc7-28b7-45a3-8070-9e87f3517dbb","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"24cf4bc7-28b7-45a3-8070-9e87f3517dbb",{}]
INFO:ocpp:CP_TEST: receive message [3,"24cf4bc7-28b7-45a3-8070-9e87f3517dbb",{}]
INFO:MockSystemTest:3.3 Status (Plug): Preparing
INFO:ocpp:CP_TEST: send [2,"78b9abd2-d705-40ff-8109-21156274934f","Authorize",{"idTag":"RFID_TAG_1"}]
INFO:ocpp:CP_TEST: receive message [2,"78b9abd2-d705-40ff-8109-21156274934f","Authorize",{"idTag":"RFID_TAG_1"}]
INFO:MockSystemTest:[CSMS] <- Authorize: {'id_tag': 'RFID_TAG_1'}
INFO:MockSystemTest:[CSMS] -> AuthorizeConf: Authorize(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"78b9abd2-d705-40ff-8109-21156274934f",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"78b9abd2-d705-40ff-8109-21156274934f",{"idTagInfo":{"status":"Accepted"}}]
INFO:MockSystemTest:3.4 Authorize: Accepted
INFO:ocpp:CP_TEST: send [2,"13452001-4998-4dc9-9c05-6ba6249bec44","StartTransaction",{"connectorId":1,"idTag":"RFID_TAG_1","meterStart":0,"timestamp":"2026-01-05T15:26:44.682570"}]
INFO:ocpp:CP_TEST: receive message [2,"13452001-4998-4dc9-9c05-6ba6249bec44","StartTransaction",{"connectorId":1,"idTag":"RFID_TAG_1","meterStart":0,"timestamp":"2026-01-05T15:26:44.682570"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'RFID_TAG_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:44.682570'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"13452001-4998-4dc9-9c05-6ba6249bec44",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"13452001-4998-4dc9-9c05-6ba6249bec44",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:3.5 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"facf9c5c-ce7f-467b-957a-3b7791b02a2f","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"facf9c5c-ce7f-467b-957a-3b7791b02a2f","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
```

```
INFO:ocpp:CP_TEST: send [3,"facf9c5c-ce7f-467b-957a-3b7791b02a2f",{}]  
INFO:ocpp:CP_TEST: receive message [3,"facf9c5c-ce7f-467b-957a-3b7791b02a2f",{}]  
INFO:MockSystemTest:3.6 Status: Charging  
INFO:ocpp:CP_TEST: send [2,"6be6c0c4-4100-4e98-87e5-9c6ad28d02ed","MeterValues",{  
  "connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:46.685210","sampledValue":[{"value":"230.0","context":"Sample.Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.Periodic","format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],"transactionId":123}]  
INFO:MockSystemTest:3.7 MeterValues received.  
INFO:ocpp:CP_TEST: receive message [2,"6be6c0c4-4100-4e98-87e5-9c6ad28d02ed","MeterValues",{  
  "connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:46.685210","sampledValue":[{"value":"230.0","context":"Sample.Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.Periodic","format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],"transactionId":123}]  
INFO:ocpp:CP_TEST: send [2,"64358f98-2a1e-4ab1-9eaa-841d9fdfa3a8","StopTransaction",{  
  "meterStop":100,"timestamp":"2026-01-05T15:26:46.687019","transactionId":123,"reason":"Local"}]  
INFO:MockSystemTest:[CSMS] <- MeterValues: {'connector_id': 1, 'meter_value': [{'timestamp': '2026-01-05T15:26:46.685210', 'sampled_value': [{'value': '230.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Voltage', 'unit': 'V'}, {'value': '16.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Current.Import', 'unit': 'A'}, {'value': '3680.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Power.Active.Import', 'unit': 'W'}]}]}, 'transaction_id': 123}  
INFO:MockSystemTest:[CSMS] -> MeterValuesConf: {}  
INFO:ocpp:CP_TEST: send [3,"6be6c0c4-4100-4e98-87e5-9c6ad28d02ed",{}]  
INFO:ocpp:CP_TEST: receive message [2,"64358f98-2a1e-4ab1-9eaa-841d9fdfa3a8","StopTransaction",{  
  "meterStop":100,"timestamp":"2026-01-05T15:26:46.687019","transactionId":123,"reason":"Local"}]  
INFO:ocpp:CP_TEST: receive message [3,"6be6c0c4-4100-4e98-87e5-9c6ad28d02ed",{}]  
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:26:46.687019', 'transaction_id': 123, 'reason': 'Local'}  
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})  
ERROR:ocpp:Ignoring response with unknown unique id: <CallResult - unique_id=6be6c0c4-4100-4e98-87e5-9c6ad28d02ed, action=None, payload={}>  
INFO:ocpp:CP_TEST: send [3,"64358f98-2a1e-4ab1-9eaa-841d9fdfa3a8",{  
  "idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp:CP_TEST: receive message [3,"64358f98-2a1e-4ab1-9eaa-841d9fdfa3a8",{  
  "idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp_16_interface:Transaction stopped on 1: 123  
INFO:MockSystemTest:HARDWARE: Charger stopped.  
INFO:MockSystemTest:3.8 StopTransaction: Sent  
INFO:ocpp:CP_TEST: send [2,"cbfb301d-9270-4214-b70d-8a4e5a5d78f9","StatusNotification",{  
  "connectorId":1,"errorCode":"NoError","status":"Finishing"}]  
INFO:ocpp:CP_TEST: receive message [2,"cbfb301d-9270-4214-b70d-8a4e5a5d78f9","StatusNotification",{  
  "connectorId":1,"errorCode":"NoError","status":"Finishing"}]  
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}  
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}  
INFO:ocpp:CP_TEST: send [3,"cbfb301d-9270-4214-b70d-8a4e5a5d78f9",{}]  
INFO:ocpp:CP_TEST: receive message [3,"cbfb301d-9270-4214-b70d-8a4e5a5d78f9",{}]  
INFO:MockSystemTest:3.9 Status: Finishing  
INFO:ocpp:CP_TEST: send [2,"e65605de-59e2-4ca6-a2bb-65103785421e","StatusNotification",{  
  "connectorId":1,"errorCode":"NoError","status":"Available"}]  
INFO:ocpp:CP_TEST: receive message [2,"e65605de-59e2-4ca6-a2bb-65103785421e","StatusNotification",{  
  "connectorId":1,"errorCode":"NoError","status":"Available"}]  
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}  
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}  
INFO:ocpp:CP_TEST: send [3,"e65605de-59e2-4ca6-a2bb-65103785421e",{}]  
INFO:ocpp:CP_TEST: receive message [3,"e65605de-59e2-4ca6-a2bb-65103785421e",{}]  
INFO:MockSystemTest:3.10 Status: Available  
INFO:ocpp:CP_TEST: send [2,"8b3597ac-d3c1-4268-b358-6354b61b4f69","StatusNotification",{  
  "connectorId":1,"errorCode":"NoError","status":"Preparing"}]  
INFO:ocpp:CP_TEST: receive message [2,"8b3597ac-d3c1-4268-b358-6354b61b4f69","StatusNotification",{  
  "connectorId":1,"errorCode":"NoError","status":"Preparing"}]  
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}  
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}  
INFO:ocpp:CP_TEST: send [3,"8b3597ac-d3c1-4268-b358-6354b61b4f69",{}]  
INFO:ocpp:CP_TEST: receive message [3,"8b3597ac-d3c1-4268-b358-6354b61b4f69",{}]  
INFO:MockSystemTest:3.11 Status: Preparing  
INFO:ocpp:CP_TEST: send [2,"9de0efee-9d14-4f22-a5d2-b17fc0535a87","RemoteStartTransaction",{  
  "idTag":"REMOTE_TAG_1"}]  
INFO:ocpp:CP_TEST: receive message [2,"9de0efee-9d14-4f22-a5d2-b17fc0535a87","RemoteStartTransaction",{  
  "idTag":"REMOTE_TAG_1"}]  
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: REMOTE_TAG_1  
INFO:ocpp:CP_TEST: send [2,"603b5a31-e330-473c-9a47-d086a1398623","StartTransaction",{  
  "connectorId":1,"idTag":"REMOTE_TAG_1","meterStart":0,"timestamp":"2026-01-05T15:26:46.696473"}]  
INFO:ocpp:CP_TEST: receive message [2,"603b5a31-e330-473c-9a47-d086a1398623","StartTransaction",{  
  "connectorId":1,"idTag":"REMOTE_TAG_1","meterStart":0,"timestamp":"2026-01-05T15:26:46.696473"}]  
INFO:ocpp:CP_TEST: send [3,"9de0efee-9d14-4f22-a5d2-b17fc0535a87",{  
  "status":"Accepted"}]  
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'REMOTE_TAG_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:46.696473'}  
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})  
INFO:ocpp:CP_TEST: send [3,"603b5a31-e330-473c-9a47-d086a1398623",{  
  "transactionId":123,"idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp:CP_TEST: receive message [3,"9de0efee-9d14-4f22-a5d2-b17fc0535a87",{  
  "status":"Accepted"}]  
INFO:ocpp:CP_TEST: receive message [3,"603b5a31-e330-473c-9a47-d086a1398623",{  
  "transactionId":123,"idTagInfo":{"status":"Accepted"}}]  
INFO:MockSystemTest:3.12 RemoteStartTransaction: Accepted  
INFO:ocpp:CP_TEST: send [2,"3d3fd08-00e9-45f3-89c3-05cdd47c8b4e","StartTransaction",{  
  "connectorId":1,"idTag":"REMOTE_TAG_1","meterStart":0,"timestamp":"2026-01-05T15:26:46.698683"}]
```

```
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2,"3d3f9d08-00e9-45f3-89c3-05cdd47c8b4e","StartTransaction",{"connectorId":1,"idTag":"REMOTE_TAG_1","meterStart":0,"timestamp":"2026-01-05T15:26:46.698683"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'REMOTE_TAG_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:46.698683'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"3d3f9d08-00e9-45f3-89c3-05cdd47c8b4e",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"3d3f9d08-00e9-45f3-89c3-05cdd47c8b4e",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:3.13 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"647ce91a-825d-40ac-88ed-0e7b4d4dbaf4","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"647ce91a-825d-40ac-88ed-0e7b4d4dbaf4","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"647ce91a-825d-40ac-88ed-0e7b4d4dbaf4",{}]
INFO:ocpp:CP_TEST: receive message [3,"647ce91a-825d-40ac-88ed-0e7b4d4dbaf4",{}]
INFO:MockSystemTest:3.14 Status: Charging
INFO:ocpp:CP_TEST: send [2,"46d4571a-b710-4fe2-94aa-daf3f3d78c87","MeterValues",{"connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:48.699755","sampledValue":[{"value":"230.0","context":"Sample.Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.Periodic","format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],{"transactionId":123}]
INFO:ocpp:CP_TEST: receive message [2,"46d4571a-b710-4fe2-94aa-daf3f3d78c87","MeterValues",{"connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:48.699755","sampledValue":[{"value":"230.0","context":"Sample.Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.Periodic","format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],{"transactionId":123}]
INFO:MockSystemTest:3.15 MeterValues received.
INFO:MockSystemTest:[CSMS] <- MeterValues: {'connector_id': 1, 'meter_value': [{'timestamp': '2026-01-05T15:26:48.699755', 'sampled_value': [{'value': '230.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Voltage', 'unit': 'V'}, {'value': '16.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Current.Import', 'unit': 'A'}, {'value': '3680.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Power.Active.Import', 'unit': 'W'}]}], 'transaction_id': 123}
INFO:MockSystemTest:[CSMS] -> MeterValuesConf: {}
INFO:ocpp:CP_TEST: send [2,"5f7ca339-c1ad-4378-b572-eedd0b3cde5b","RemoteStopTransaction",{"transactionId":123}]
INFO:ocpp:CP_TEST: send [3,"46d4571a-b710-4fe2-94aa-daf3f3d78c87",{}]
INFO:ocpp:CP_TEST: receive message [2,"5f7ca339-c1ad-4378-b572-eedd0b3cde5b","RemoteStopTransaction",{"transactionId":123}]
INFO:ocpp_16_interface:Received RemoteStopTransaction for transaction_id: 123
INFO:ocpp:CP_TEST: send [3,"5f7ca339-c1ad-4378-b572-eedd0b3cde5b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"46d4571a-b710-4fe2-94aa-daf3f3d78c87",{}]
INFO:ocpp:CP_TEST: receive message [3,"5f7ca339-c1ad-4378-b572-eedd0b3cde5b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"3e7666f4-30f7-4850-9dc5-5247eb308161","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:26:48.705688","transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:3.16 RemoteStopTransaction: Accepted
INFO:ocpp:CP_TEST: receive message [2,"3e7666f4-30f7-4850-9dc5-5247eb308161","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:26:48.705688","transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:26:48.705688', 'transaction_id': 123, 'reason': 'Remote'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"3e7666f4-30f7-4850-9dc5-5247eb308161",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"3e7666f4-30f7-4850-9dc5-5247eb308161",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
WARNING:ocpp_16_interface:No transaction to stop for connector 1.
INFO:MockSystemTest:3.17 StopTransaction: Triggered by RemoteStop
INFO:ocpp:CP_TEST: send [2,"4637ec05-c75f-4be7-bece-d68a1683b2ae","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:ocpp:CP_TEST: receive message [2,"4637ec05-c75f-4be7-bece-d68a1683b2ae","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"4637ec05-c75f-4be7-bece-d68a1683b2ae",{}]
INFO:ocpp:CP_TEST: receive message [3,"4637ec05-c75f-4be7-bece-d68a1683b2ae",{}]
INFO:MockSystemTest:3.18 Status: Finishing
INFO:ocpp:CP_TEST: send [2,"d91f3ace-d7ee-416a-af54-403323e7db85","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"d91f3ace-d7ee-416a-af54-403323e7db85","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"d91f3ace-d7ee-416a-af54-403323e7db85",{}]
INFO:ocpp:CP_TEST: receive message [3,"d91f3ace-d7ee-416a-af54-403323e7db85",{}]
INFO:MockSystemTest:3.19 Status: Available
```

```
INFO:MockSystemTest:
--- 4. Reset Verification (Full Sequence) ---
INFO:ocpp:CP_TEST: send [2,"e066acbd-57a3-4863-8b0d-6a408ec93ff7","Reset",{"type":"Hard"}]
INFO:ocpp:CP_TEST: receive message [2,"e066acbd-57a3-4863-8b0d-6a408ec93ff7","Reset",{"type":"Hard"}]
INFO:ocpp_16_interface:Received Reset type: Hard
INFO:ocpp:CP_TEST: send [3,"e066acbd-57a3-4863-8b0d-6a408ec93ff7",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"e066acbd-57a3-4863-8b0d-6a408ec93ff7",{"status":"Accepted"}]
INFO:MockSystemTest:4.1 Reset (Hard): Accepted
INFO:MockSystemTest:4.2 Waiting for BootNotification...
INFO:ocpp:CP_TEST: send [2,"c08d2559-4d0a-47dd-8244-2ec72ff48f4f","MeterValues",{"connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:50.706806","sampledValue":[{"value":"230.0","context":"Sample.Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.Periodic","format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],"transactionId":123}]
INFO:ocpp:CP_TEST: receive message [2,"c08d2559-4d0a-47dd-8244-2ec72ff48f4f","MeterValues",{"connectorId":1,"meterValue":[{"timestamp":"2026-01-05T15:26:50.706806","sampledValue":[{"value":"230.0","context":"Sample.Periodic","format":"Raw","measurand":"Voltage","unit":"V"},{"value":"16.0","context":"Sample.Periodic","format":"Raw","measurand":"Current.Import","unit":"A"},{"value":"3680.0","context":"Sample.Periodic","format":"Raw","measurand":"Power.Active.Import","unit":"W"}]}],"transactionId":123}]
INFO:MockSystemTest:[CSMS] <- MeterValues: {'connector_id': 1, 'meter_value': [{'timestamp': '2026-01-05T15:26:50.706806', 'sampled_value': [{'value': '230.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Voltage', 'unit': 'V'}, {'value': '16.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Current.Import', 'unit': 'A'}, {'value': '3680.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Power.Active.Import', 'unit': 'W'}]}], 'transaction_id': 123}
INFO:MockSystemTest:[CSMS] -> MeterValuesConf: {}
INFO:ocpp:CP_TEST: send [3,"c08d2559-4d0a-47dd-8244-2ec72ff48f4f",{}]
INFO:ocpp:CP_TEST: receive message [3,"c08d2559-4d0a-47dd-8244-2ec72ff48f4f",{}]
INFO:ocpp:CP_TEST: send [2,"0bfb95b0-0828-4940-9388-34ebbf74aee4","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2,"0bfb95b0-0828-4940-9388-34ebbf74aee4","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:26:51.715860', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3,"0bfb95b0-0828-4940-9388-34ebbf74aee4",{"currentTime":"2026-01-05T15:26:51.715860","interval":10,"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"0bfb95b0-0828-4940-9388-34ebbf74aee4",{"currentTime":"2026-01-05T15:26:51.715860","interval":10,"status":"Accepted"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:4.2 BootNotification: Sent
INFO:ocpp:CP_TEST: send [2,"2824913d-73eb-4c09-ac1d-032aeb5dc3a9","Heartbeat",{}]
INFO:ocpp:CP_TEST: receive message [2,"2824913d-73eb-4c09-ac1d-032aeb5dc3a9","Heartbeat",{}]
INFO:MockSystemTest:[CSMS] <- Heartbeat: {}
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:60: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    current_time=datetime.utcnow().isoformat()
INFO:MockSystemTest:[CSMS] -> HeartbeatConf: Heartbeat(current_time='2026-01-05T15:26:51.717832')
INFO:ocpp:CP_TEST: send [3,"2824913d-73eb-4c09-ac1d-032aeb5dc3a9",{"currentTime":"2026-01-05T15:26:51.717832"}]
INFO:ocpp:CP_TEST: receive message [3,"2824913d-73eb-4c09-ac1d-032aeb5dc3a9",{"currentTime":"2026-01-05T15:26:51.717832"}]
INFO:MockSystemTest:4.3 Heartbeat: Sent
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:435: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    await cp.call(call.MeterValues(connector_id=0, meter_value=[{"timestamp": datetime.utcnow().isoformat(), "sampledValue": [{"value": "0"}]}]))
INFO:ocpp:CP_TEST: send [2,"d521ffbc-f0f3-4ac2-b13e-c932117fbc9c","MeterValues",{"connectorId":0,"meterValue":[{"timestamp":"2026-01-05T15:26:51.718879","sampledValue":[{"value":"0"}]}]}]
INFO:ocpp:CP_TEST: receive message [2,"d521ffbc-f0f3-4ac2-b13e-c932117fbc9c","MeterValues",{"connectorId":0,"meterValue":[{"timestamp":"2026-01-05T15:26:51.718879","sampledValue":[{"value":"0"}]}]}]
INFO:MockSystemTest:[CSMS] <- MeterValues: {'connector_id': 0, 'meter_value': [{'timestamp': '2026-01-05T15:26:51.718879', 'sampled_value': [{'value': '0'}]}]}
INFO:MockSystemTest:[CSMS] -> MeterValuesConf: {}
INFO:ocpp:CP_TEST: send [3,"d521ffbc-f0f3-4ac2-b13e-c932117fbc9c",{}]
INFO:ocpp:CP_TEST: receive message [3,"d521ffbc-f0f3-4ac2-b13e-c932117fbc9c",{}]
INFO:MockSystemTest:4.4 MeterValues: Sent
INFO:ocpp:CP_TEST: send [2,"a6ae2fdd-2967-434c-885c-b9786247b2dc","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"a6ae2fdd-2967-434c-885c-b9786247b2dc","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"a6ae2fdd-2967-434c-885c-b9786247b2dc",{}]
INFO:ocpp:CP_TEST: receive message [3,"a6ae2fdd-2967-434c-885c-b9786247b2dc",{}]
INFO:MockSystemTest:4.5 StatusNotification: Available
INFO:ocpp:CP_TEST: send [2,"98c13de0-9658-4261-84fb-4610c816b642","ChangeConfiguration",{"key":"HeartbeatInterval","value":"600"}]
INFO:ocpp:CP_TEST: receive message [2,"98c13de0-9658-4261-84fb-4610c816b642","ChangeConfiguration",{"key":"HeartbeatInterval","value":"600"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for HeartbeatInterval to 600
```



```
INFO:ocpp:CP_TEST: send [3,"98c13de0-9658-4261-84fb-4610c816b642",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"98c13de0-9658-4261-84fb-4610c816b642",{"status":"Accepted"}]
INFO:MockSystemTest:4.6 ChangeConfiguration (Heartbeat): Accepted
INFO:ocpp:CP_TEST: send [2,"5603c4db-6f19-46c8-aac6-45269515573f", "ChangeConfiguration",{"key":"MeterValueSampleInterval", "value":"60"}]
INFO:ocpp:CP_TEST: receive message [2,"5603c4db-6f19-46c8-aac6-45269515573f", "ChangeConfiguration",{"key":"MeterValueSampleInterval", "value":"60"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for MeterValueSampleInterval to 60
INFO:ocpp:CP_TEST: send [3,"5603c4db-6f19-46c8-aac6-45269515573f",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"5603c4db-6f19-46c8-aac6-45269515573f",{"status":"Accepted"}]
INFO:MockSystemTest:4.7 ChangeConfiguration (MeterValue): Accepted
INFO:ocpp:CP_TEST: send [2,"0816fbb3-0ca2-4a13-a148-aaba89ced3eb", "ChangeConfiguration",{"key":"LocalAuthorizeOffline", "value":"True"}]
INFO:ocpp:CP_TEST: receive message [2,"0816fbb3-0ca2-4a13-a148-aaba89ced3eb", "ChangeConfiguration",{"key":"LocalAuthorizeOffline", "value":"True"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for LocalAuthorizeOffline to True
INFO:ocpp:CP_TEST: send [3,"0816fbb3-0ca2-4a13-a148-aaba89ced3eb",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"0816fbb3-0ca2-4a13-a148-aaba89ced3eb",{"status":"Accepted"}]
INFO:MockSystemTest:4.8 ChangeConfiguration (LocalAuth): Accepted
INFO:ocpp:CP_TEST: send [2,"7da2466e-9cd0-4e12-be93-22e3184f192a", "Reset", {"type":"Soft"}]
INFO:ocpp:CP_TEST: receive message [2,"7da2466e-9cd0-4e12-be93-22e3184f192a", "Reset", {"type":"Soft"}]
INFO:ocpp_16_interface:Received Reset type: Soft
INFO:ocpp:CP_TEST: send [3,"7da2466e-9cd0-4e12-be93-22e3184f192a", {"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"7da2466e-9cd0-4e12-be93-22e3184f192a", {"status":"Accepted"}]
INFO:MockSystemTest:4.9 Reset (Soft): Accepted
INFO:ocpp:CP_TEST: send [2,"1e6fd621-34ff-45e0-9272-86cbb5208786", "MeterValues", {"connectorId":1, "meterValue": [{"timestamp": "2026-01-05T15:26:52.710944", "sampledValue": [{"value": "230.0", "context": "Sample.Periodic", "format": "Raw", "measurand": "Voltage", "unit": "V"}, {"value": "16.0", "context": "Sample.Periodic", "format": "Raw", "measurand": "Current.Import", "unit": "A"}, {"value": "3680.0", "context": "Sample.Periodic", "format": "Raw", "measurand": "Power.Active.Import", "unit": "W"}]}], "transactionId":123}]
INFO:ocpp:CP_TEST: receive message [2,"1e6fd621-34ff-45e0-9272-86cbb5208786", "MeterValues", {"connectorId":1, "meterValue": [{"timestamp": "2026-01-05T15:26:52.710944", "sampledValue": [{"value": "230.0", "context": "Sample.Periodic", "format": "Raw", "measurand": "Voltage", "unit": "V"}, {"value": "16.0", "context": "Sample.Periodic", "format": "Raw", "measurand": "Current.Import", "unit": "A"}, {"value": "3680.0", "context": "Sample.Periodic", "format": "Raw", "measurand": "Power.Active.Import", "unit": "W"}]}], "transactionId":123}]
INFO:MockSystemTest:[CSMS] <- MeterValues: {'connector_id': 1, 'meter_value': [{'timestamp': '2026-01-05T15:26:52.710944', 'sampled_value': [{'value': '230.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Voltage', 'unit': 'V'}, {'value': '16.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Current.Import', 'unit': 'A'}, {'value': '3680.0', 'context': 'Sample.Periodic', 'format': 'Raw', 'measurand': 'Power.Active.Import', 'unit': 'W'}]}], 'transaction_id': 123}
INFO:MockSystemTest:[CSMS] -> MeterValuesConf: {}
INFO:ocpp:CP_TEST: send [3,"1e6fd621-34ff-45e0-9272-86cbb5208786",{}]]
INFO:ocpp:CP_TEST: receive message [3,"1e6fd621-34ff-45e0-9272-86cbb5208786",{}]]
INFO:ocpp:CP_TEST: send [2,"ea55ac68-b7c8-4454-8056-4272f83b6a75", "BootNotification", {"chargePointModel": "MEA-V2G-01", "chargePointVendor": "KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2,"ea55ac68-b7c8-4454-8056-4272f83b6a75", "BootNotification", {"chargePointModel": "MEA-V2G-01", "chargePointVendor": "KMUTNB"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:26:52.730265', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3,"ea55ac68-b7c8-4454-8056-4272f83b6a75", {"currentTime": "2026-01-05T15:26:52.730265", "interval": 10, "status": "Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"ea55ac68-b7c8-4454-8056-4272f83b6a75", {"currentTime": "2026-01-05T15:26:52.730265", "interval": 10, "status": "Accepted"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:4.10 BootNotification: Sent
INFO:ocpp:CP_TEST: send [2,"d61134fc-3cd9-473c-9591-2f40cdc32bc0", "Heartbeat", {}]
INFO:ocpp:CP_TEST: receive message [2,"d61134fc-3cd9-473c-9591-2f40cdc32bc0", "Heartbeat", {}]
INFO:MockSystemTest:[CSMS] <- Heartbeat: {}
INFO:MockSystemTest:[CSMS] -> HeartbeatConf: Heartbeat(current_time='2026-01-05T15:26:52.732645')
INFO:ocpp:CP_TEST: send [3,"d61134fc-3cd9-473c-9591-2f40cdc32bc0", {"currentTime": "2026-01-05T15:26:52.732645"}]
INFO:ocpp:CP_TEST: receive message [3,"d61134fc-3cd9-473c-9591-2f40cdc32bc0", {"currentTime": "2026-01-05T15:26:52.732645"}]
INFO:MockSystemTest:4.11 Heartbeat: Sent
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:468: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    await cp.call(call.MeterValues(connector_id=0, meter_value=[{"timestamp": datetime.utcnow().isoformat(), "sampledValue": [{"value": "0"}]}]))
INFO:ocpp:CP_TEST: send [2,"1f394128-e1de-43f6-aefc-584767ff4096", "MeterValues", {"connectorId":0, "meterValue": [{"timestamp": "2026-01-05T15:26:52.733677", "sampledValue": [{"value": "0"}]}]}]
INFO:ocpp:CP_TEST: receive message [2,"1f394128-e1de-43f6-aefc-584767ff4096", "MeterValues", {"connectorId":0, "meterValue": [{"timestamp": "2026-01-05T15:26:52.733677", "sampledValue": [{"value": "0"}]}]}]
INFO:MockSystemTest:[CSMS] <- MeterValues: {'connector_id': 0, 'meter_value': [{'timestamp': '2026-01-05T15:26:52.733677', 'sampled_value': [{'value': '0'}]}]}
INFO:MockSystemTest:[CSMS] -> MeterValuesConf: {}
INFO:ocpp:CP_TEST: send [3,"1f394128-e1de-43f6-aefc-584767ff4096",{}]]
INFO:ocpp:CP_TEST: receive message [3,"1f394128-e1de-43f6-aefc-584767ff4096",{}]]
INFO:MockSystemTest:4.12 MeterValues: Sent
INFO:ocpp:CP_TEST: send [2,"c4370b79-20c2-4f76-ac4c-d19763d5c95e", "StatusNotification", {"connectorId":1, "errorCode": "NoError", "status": "Available"}]
```

```
INFO:ocpp:CP_TEST: receive message [2,"c4370b79-20c2-4f76-ac4c-d19763d5c95e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"c4370b79-20c2-4f76-ac4c-d19763d5c95e",{}]
INFO:ocpp:CP_TEST: receive message [3,"c4370b79-20c2-4f76-ac4c-d19763d5c95e",{}]
INFO:MockSystemTest:4.13 StatusNotification: Available
INFO:ocpp:CP_TEST: send [2,"0619a04f-eb38-4b93-9507-3203a7aaed6e","ChangeConfiguration",{"key":"HeartbeatInterval","value":"600"}]
INFO:ocpp:CP_TEST: receive message [2,"0619a04f-eb38-4b93-9507-3203a7aaed6e","ChangeConfiguration",{"key":"HeartbeatInterval","value":"600"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for HeartbeatInterval to 600
INFO:ocpp:CP_TEST: send [3,"0619a04f-eb38-4b93-9507-3203a7aaed6e",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"0619a04f-eb38-4b93-9507-3203a7aaed6e",{"status":"Accepted"}]
INFO:MockSystemTest:4.14 ChangeConfiguration (Heartbeat): Accepted
INFO:ocpp:CP_TEST: send [2,"ce3a3325-7007-4bd8-bb28-d1edad81527a","ChangeConfiguration",{"key":"MeterValueSampleInterval","value":"60"}]
INFO:ocpp:CP_TEST: receive message [2,"ce3a3325-7007-4bd8-bb28-d1edad81527a","ChangeConfiguration",{"key":"MeterValueSampleInterval","value":"60"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for MeterValueSampleInterval to 60
INFO:ocpp:CP_TEST: send [3,"ce3a3325-7007-4bd8-bb28-d1edad81527a",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"ce3a3325-7007-4bd8-bb28-d1edad81527a",{"status":"Accepted"}]
INFO:MockSystemTest:4.15 ChangeConfiguration (MeterValue): Accepted
INFO:ocpp:CP_TEST: send [2,"f8016eb6-617d-4405-b69d-3b52e67c78a2","ChangeConfiguration",{"key":"LocalAuthorizeOffline","value":"True"}]
INFO:ocpp:CP_TEST: receive message [2,"f8016eb6-617d-4405-b69d-3b52e67c78a2","ChangeConfiguration",{"key":"LocalAuthorizeOffline","value":"True"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for LocalAuthorizeOffline to True
INFO:ocpp:CP_TEST: send [3,"f8016eb6-617d-4405-b69d-3b52e67c78a2",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"f8016eb6-617d-4405-b69d-3b52e67c78a2",{"status":"Accepted"}]
INFO:MockSystemTest:4.16 ChangeConfiguration (LocalAuth): Accepted
INFO:ocpp:CP_TEST: send [2,"99241ac4-74b6-4764-a610-fcd1e640f55e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"99241ac4-74b6-4764-a610-fcd1e640f55e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"99241ac4-74b6-4764-a610-fcd1e640f55e",{}]
INFO:ocpp:CP_TEST: receive message [3,"99241ac4-74b6-4764-a610-fcd1e640f55e",{}]
INFO:MockSystemTest:4.17 StatusNotification: Available
INFO:ocpp:CP_TEST: send [2,"cd38f198-5128-4e77-8e72-c98262b952c2","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"cd38f198-5128-4e77-8e72-c98262b952c2","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"cd38f198-5128-4e77-8e72-c98262b952c2",{}]
INFO:ocpp:CP_TEST: receive message [3,"cd38f198-5128-4e77-8e72-c98262b952c2",{}]
INFO:ocpp:CP_TEST: send [2,"a3dcf4eb-cad0-4dfb-998d-85c348d3e7f8","StartTransaction",{"connectorId":1,"idTag":"RESET_TEST_TAG","meterStart":0,"timestamp":"2026-01-05T15:26:52.745489"}]
INFO:ocpp:CP_TEST: receive message [2,"a3dcf4eb-cad0-4dfb-998d-85c348d3e7f8","StartTransaction",{"connectorId":1,"idTag":"RESET_TEST_TAG","meterStart":0,"timestamp":"2026-01-05T15:26:52.745489"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'RESET_TEST_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:52.745489'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"a3dcf4eb-cad0-4dfb-998d-85c348d3e7f8",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"a3dcf4eb-cad0-4dfb-998d-85c348d3e7f8",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:4.18 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"cdd19602-b163-4a8a-b4e7-a9a5f5d5ee6a","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"cdd19602-b163-4a8a-b4e7-a9a5f5d5ee6a","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"cdd19602-b163-4a8a-b4e7-a9a5f5d5ee6a",{}]
INFO:ocpp:CP_TEST: receive message [3,"cdd19602-b163-4a8a-b4e7-a9a5f5d5ee6a",{}]
INFO:ocpp:CP_TEST: send [2,"34de74f1-7254-4dea-aa8b-552da0b5cc3f","Reset",{"type":"Hard"}]
INFO:ocpp:CP_TEST: receive message [2,"34de74f1-7254-4dea-aa8b-552da0b5cc3f","Reset",{"type":"Hard"}]
INFO:ocpp_16_interface:Received Reset type: Hard
INFO:ocpp:CP_TEST: send [3,"34de74f1-7254-4dea-aa8b-552da0b5cc3f",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"34de74f1-7254-4dea-aa8b-552da0b5cc3f",{"status":"Accepted"}]
INFO:MockSystemTest:4.19 Reset (Hard): Accepted
INFO:ocpp:CP_TEST: send [2,"4c913220-230a-4390-bcc2-93feb64603f3","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:26:52.750476","transactionId":123,"reason":"HardReset"}]
INFO:ocpp:CP_TEST: receive message [2,"4c913220-230a-4390-bcc2-93feb64603f3","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:26:52.750476","transactionId":123,"reason":"HardReset"}]
```

```

INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:26:52.750476', 'transaction_id': 123, 'reason': 'HardReset'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3, "4c913220-230a-4390-bcc2-93feb64603f3", {"idTagInfo": {"status": "Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3, "4c913220-230a-4390-bcc2-93feb64603f3", {"idTagInfo": {"status": "Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:4.20 StopTransaction: Sent
INFO:ocpp:CP_TEST: send [2, "1e38723c-353e-4019-9759-5f87d9560dfa", "BootNotification", {"chargePointModel": "MEA-V2G-01", "chargePointVendor": "KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2, "1e38723c-353e-4019-9759-5f87d9560dfa", "BootNotification", {"chargePointModel": "MEA-V2G-01", "chargePointVendor": "KMUTNB"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:26:53.755359', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3, "1e38723c-353e-4019-9759-5f87d9560dfa", {"currentTime": "2026-01-05T15:26:53.755359", "interval": 10, "status": "Accepted"}]
INFO:ocpp:CP_TEST: receive message [3, "1e38723c-353e-4019-9759-5f87d9560dfa", {"currentTime": "2026-01-05T15:26:53.755359", "interval": 10, "status": "Accepted"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:4.21 BootNotification: Sent
INFO:ocpp:CP_TEST: send [2, "84bdb916-0802-4865-a2a8-4f78e488db67", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:ocpp:CP_TEST: receive message [2, "84bdb916-0802-4865-a2a8-4f78e488db67", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "84bdb916-0802-4865-a2a8-4f78e488db67", {}]
INFO:ocpp:CP_TEST: receive message [3, "84bdb916-0802-4865-a2a8-4f78e488db67", {}]
INFO:MockSystemTest:4.21 StatusNotification (Unplug): Available
INFO:ocpp:CP_TEST: send [2, "d3947ede-96e2-437e-869d-69385b36cbbb", "BootNotification", {"chargePointModel": "MEA-V2G-01", "chargePointVendor": "KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2, "d3947ede-96e2-437e-869d-69385b36cbbb", "BootNotification", {"chargePointModel": "MEA-V2G-01", "chargePointVendor": "KMUTNB"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:26:53.759553', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3, "d3947ede-96e2-437e-869d-69385b36cbbb", {"currentTime": "2026-01-05T15:26:53.759553", "interval": 10, "status": "Accepted"}]
INFO:ocpp:CP_TEST: receive message [3, "d3947ede-96e2-437e-869d-69385b36cbbb", {"currentTime": "2026-01-05T15:26:53.759553", "interval": 10, "status": "Accepted"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:
--- 5. Reservation Verification (Full Sequence) ---
INFO:ocpp:CP_TEST: send [2, "8801e936-8e40-43c0-940d-76ec81415b14", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:ocpp:CP_TEST: receive message [2, "8801e936-8e40-43c0-940d-76ec81415b14", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "8801e936-8e40-43c0-940d-76ec81415b14", {}]
INFO:ocpp:CP_TEST: receive message [3, "8801e936-8e40-43c0-940d-76ec81415b14", {}]
INFO:MockSystemTest:5.1 Status: Available
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:526: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to
    represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    expiry = (datetime.utcnow().replace(microsecond=0).timestamp() + 3600)
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:529: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects
    to represent datetimes in UTC: datetime.datetime.fromtimestamp(timestamp, datetime.UTC).
    expiry_date=datetime.datetime.fromtimestamp(expiry).isoformat(),
INFO:ocpp:CP_TEST: send [2, "66d8dd07-478f-4d37-aeaf-20301a7c051f", "ReserveNow", {"connectorId": 1, "expiryDate": "2026-01-05T09:26:53", "idTag": "RES_TAG", "reservationId": 1}]
INFO:ocpp:CP_TEST: receive message [2, "66d8dd07-478f-4d37-aeaf-20301a7c051f", "ReserveNow", {"connectorId": 1, "expiryDate": "2026-01-05T09:26:53", "idTag": "RES_TAG", "reservationId": 1}]
INFO:ocpp_16_interface:Received ReserveNow for RES_TAG
INFO:ocpp:CP_TEST: send [3, "66d8dd07-478f-4d37-aeaf-20301a7c051f", {"status": "Accepted"}]
INFO:ocpp:CP_TEST: receive message [3, "66d8dd07-478f-4d37-aeaf-20301a7c051f", {"status": "Accepted"}]
INFO:MockSystemTest:5.2 ReserveNow: Accepted
INFO:ocpp:CP_TEST: send [2, "2ee5c50b-c127-4ac7-9a14-6c95bb165cc4", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Reserved"}]
INFO:ocpp:CP_TEST: receive message [2, "2ee5c50b-c127-4ac7-9a14-6c95bb165cc4", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Reserved"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Reserved'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "2ee5c50b-c127-4ac7-9a14-6c95bb165cc4", {}]
INFO:ocpp:CP_TEST: receive message [3, "2ee5c50b-c127-4ac7-9a14-6c95bb165cc4", {}]
INFO:MockSystemTest:5.3 Status: Reserved
INFO:ocpp:CP_TEST: send [2, "e43d2355-cc6e5-4282-a9bb-9c41d55e033d", "CancelReservation", {"reservationId": 1}]

```

```
INFO:ocpp:CP_TEST: receive message [2,"e43d2355-cce5-4282-a9bb-9c41d55e033d","CancelReservation",{"reservationId":1}]
INFO:ocpp_i6_interface:Received CancelReservation for 1
INFO:ocpp:CP_TEST: send [3,"e43d2355-cce5-4282-a9bb-9c41d55e033d",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"e43d2355-cce5-4282-a9bb-9c41d55e033d",{"status":"Accepted"}]
INFO:MockSystemTest:5.4 CancelReservation: Accepted
INFO:ocpp:CP_TEST: send [2,"308284fa-4e32-402d-be2e-c3e24c92b8a0","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"308284fa-4e32-402d-be2e-c3e24c92b8a0","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"308284fa-4e32-402d-be2e-c3e24c92b8a0",{}]
INFO:ocpp:CP_TEST: receive message [3,"308284fa-4e32-402d-be2e-c3e24c92b8a0",{}]
INFO:MockSystemTest:5.5 Status: Available
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:548: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to
represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    expiry_short = (datetime.datetime.utcnow().replace(microsecond=0).timestamp() + 2) # 2 seconds
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:551: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects
to represent datetimes in UTC: datetime.datetime.fromtimestamp(timestamp, datetime.UTC).
    expiry_date=datetime.datetime.fromtimestamp(expiry_short).isoformat(),
INFO:ocpp:CP_TEST: send [2,"e6978eea-efa9-4c7c-82b2-4722f4be7f3a","ReserveNow",{"connectorId":1,"expiryDate":"2026-01-05T08:26:55","idTag":"RES_TAG_EXP","reservationId":2}]
INFO:ocpp:CP_TEST: receive message [2,"e6978eea-efa9-4c7c-82b2-4722f4be7f3a","ReserveNow",{"connectorId":1,"expiryDate":"2026-01-05T08:26:55","idTag":"RES_TAG_EXP","reservationId":2}]
INFO:ocpp_i6_interface:Received ReserveNow for RES_TAG_EXP
INFO:ocpp:CP_TEST: send [3,"e6978eea-efa9-4c7c-82b2-4722f4be7f3a",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"e6978eea-efa9-4c7c-82b2-4722f4be7f3a",{"status":"Accepted"}]
INFO:MockSystemTest:5.6 ReserveNow (Short): Accepted
INFO:ocpp:CP_TEST: send [2,"ff156836-cfbf-40eb-96dd-6b19c04df350","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Reserved"}]
INFO:ocpp:CP_TEST: receive message [2,"ff156836-cfbf-40eb-96dd-6b19c04df350","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Reserved"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Reserved'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"ff156836-cfbf-40eb-96dd-6b19c04df350",{}]
INFO:ocpp:CP_TEST: receive message [3,"ff156836-cfbf-40eb-96dd-6b19c04df350",{}]
INFO:MockSystemTest:5.7 Status: Reserved
INFO:MockSystemTest:5.8 Waiting for reservation expiry...
INFO:ocpp:CP_TEST: send [2,"bc0c6f7c-9222-4c6c-8c77-8029b59e2041","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"bc0c6f7c-9222-4c6c-8c77-8029b59e2041","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"bc0c6f7c-9222-4c6c-8c77-8029b59e2041",{}]
INFO:ocpp:CP_TEST: receive message [3,"bc0c6f7c-9222-4c6c-8c77-8029b59e2041",{}]
INFO:MockSystemTest:5.9 Status: Available (Auto Revert)
/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py:572: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects
to represent datetimes in UTC: datetime.datetime.fromtimestamp(timestamp, datetime.UTC).
    expiry_date=datetime.datetime.fromtimestamp(expiry).isoformat(),
INFO:ocpp:CP_TEST: send [2,"f2183484-48e5-4703-b576-baf729044cad","ReserveNow",{"connectorId":1,"expiryDate":"2026-01-05T09:26:53","idTag":"RES_TAG_TX","reservationId":3}]
INFO:ocpp:CP_TEST: receive message [2,"f2183484-48e5-4703-b576-baf729044cad","ReserveNow",{"connectorId":1,"expiryDate":"2026-01-05T09:26:53","idTag":"RES_TAG_TX","reservationId":3}]
INFO:ocpp_i6_interface:Received ReserveNow for RES_TAG_TX
INFO:ocpp:CP_TEST: send [3,"f2183484-48e5-4703-b576-baf729044cad",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"f2183484-48e5-4703-b576-baf729044cad",{"status":"Accepted"}]
INFO:MockSystemTest:5.10 ReserveNow: Accepted
INFO:ocpp:CP_TEST: send [2,"d4d97d79-c4a8-42eb-bd61-f8c473a83c4e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"d4d97d79-c4a8-42eb-bd61-f8c473a83c4e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"d4d97d79-c4a8-42eb-bd61-f8c473a83c4e",{}]
INFO:ocpp:CP_TEST: receive message [3,"d4d97d79-c4a8-42eb-bd61-f8c473a83c4e",{}]
INFO:MockSystemTest:5.11 Status (Plug): Preparing
INFO:ocpp:CP_TEST: send [2,"21e2ede3-dab4-4173-b6dc-fb9651c99635","RemoteStartTransaction",{"idTag":"RES_TAG_TX"}]
INFO:ocpp:CP_TEST: receive message [2,"21e2ede3-dab4-4173-b6dc-fb9651c99635","RemoteStartTransaction",{"idTag":"RES_TAG_TX"}]
INFO:ocpp_i6_interface:Received RemoteStartTransaction for id_tag: RES_TAG_TX
INFO:ocpp:CP_TEST: send [3,"21e2ede3-dab4-4173-b6dc-fb9651c99635",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"ae20277d-285e-4fa3-a2e7-15a2ef2a9b86","StartTransaction",{"connectorId":1,"idTag":"RES_TAG_TX","meterStart":0,"timestamp":"2026-01-05T15:26:56.778516"}]
```

```
INFO:ocpp:CP_TEST: receive message [3,"21e2ede3-dab4-4173-b6dc-fb9651c99635",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"ae20277d-285e-4fa3-a2e7-15a2ef2a9b86",{"transactionId":1,"idTag":"RES_TAG_TX","meterStart":0,"timestamp":"2026-01-05T15:26:56.778516"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'RES_TAG_TX', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:56.778516'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:MockSystemTest:5.12 RemoteStartTransaction: Accepted
INFO:ocpp:CP_TEST: send [3,"ae20277d-285e-4fa3-a2e7-15a2ef2a9b86",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"ae20277d-285e-4fa3-a2e7-15a2ef2a9b86",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"id149776-c66e-4d7e-8bef-621437c00137",{"transactionId":1,"idTag":"RES_TAG_TX","meterStart":0,"timestamp":"2026-01-05T15:26:56.780415"}]
INFO:ocpp:CP_TEST: receive message [2,"id149776-c66e-4d7e-8bef-621437c00137",{"transactionId":1,"idTag":"RES_TAG_TX","meterStart":0,"timestamp":"2026-01-05T15:26:56.780415"}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'RES_TAG_TX', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:56.780415'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"id149776-c66e-4d7e-8bef-621437c00137",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"id149776-c66e-4d7e-8bef-621437c00137",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:5.13 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"3d08a659-f7a1-4fdc-9d1b-f5ad96dd53f5",{"transactionId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"3d08a659-f7a1-4fdc-9d1b-f5ad96dd53f5",{"transactionId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"3d08a659-f7a1-4fdc-9d1b-f5ad96dd53f5",{}]]
INFO:ocpp:CP_TEST: receive message [3,"3d08a659-f7a1-4fdc-9d1b-f5ad96dd53f5",{}]]
INFO:MockSystemTest:5.14 Status: Charging
INFO:MockSystemTest:5.15 MeterValues received (assumed).
INFO:ocpp:CP_TEST: send [2,"779d3b39-4928-448b-9616-f13c1cab05dd",{"transactionId":123}]
INFO:ocpp:CP_TEST: receive message [2,"779d3b39-4928-448b-9616-f13c1cab05dd",{"transactionId":123}]
INFO:ocpp_16_interface:Received RemoteStopTransaction for transaction_id: 123
INFO:ocpp:CP_TEST: send [3,"779d3b39-4928-448b-9616-f13c1cab05dd",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"779d3b39-4928-448b-9616-f13c1cab05dd",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"e2da2be3-b811-440e-ab7c-7996248a7b1d",{"transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:5.16 RemoteStopTransaction: Accepted
INFO:ocpp:CP_TEST: receive message [2,"e2da2be3-b811-440e-ab7c-7996248a7b1d",{"transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:26:58.788073', 'transaction_id': 123, 'reason': 'Remote'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"e2da2be3-b811-440e-ab7c-7996248a7b1d",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"e2da2be3-b811-440e-ab7c-7996248a7b1d",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:5.17 StopTransaction: Triggered by RemoteStop
INFO:ocpp:CP_TEST: send [2,"d47a1f6a-c8dc-4c11-87e9-dd211d819322",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:ocpp:CP_TEST: receive message [2,"d47a1f6a-c8dc-4c11-87e9-dd211d819322",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"d47a1f6a-c8dc-4c11-87e9-dd211d819322",{}]]
INFO:ocpp:CP_TEST: receive message [3,"d47a1f6a-c8dc-4c11-87e9-dd211d819322",{}]]
INFO:MockSystemTest:5.18 StatusNotification: Finishing
INFO:ocpp:CP_TEST: send [2,"5cddb346-7264-48ed-ae36-da4ec4c46a83",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"5cddb346-7264-48ed-ae36-da4ec4c46a83",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"5cddb346-7264-48ed-ae36-da4ec4c46a83",{}]]
INFO:ocpp:CP_TEST: receive message [3,"5cddb346-7264-48ed-ae36-da4ec4c46a83",{}]]
INFO:MockSystemTest:5.19 StatusNotification: Available
INFO:MockSystemTest:
--- 6. Charging Profile Verification (Full Sequence) ---
INFO:ocpp:CP_TEST: send [2,"7063c006-4221-4b10-8146-d0661059c489",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"7063c006-4221-4b10-8146-d0661059c489",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
```

```
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"7063c006-4221-4b10-8146-d0661059c489",{}]
INFO:ocpp:CP_TEST: receive message [3,"7063c006-4221-4b10-8146-d0661059c489",{}]
INFO:MockSystemTest:6.1 StatusNotification (Unplug): Available
INFO:ocpp:CP_TEST: send [2,"12a2da62-83d6-4b33-9976-d3f759952a3a","StatusNotification",{ "connectorId":1,"errorCode":"NoError","status":"Preparing"}}
INFO:ocpp:CP_TEST: receive message [2,"12a2da62-83d6-4b33-9976-d3f759952a3a","StatusNotification",{ "connectorId":1,"errorCode":"NoError","status":"Preparing"}}
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"12a2da62-83d6-4b33-9976-d3f759952a3a",{}]
INFO:ocpp:CP_TEST: receive message [3,"12a2da62-83d6-4b33-9976-d3f759952a3a",{}]
INFO:MockSystemTest:6.2 StatusNotification (Plug): Preparing
INFO:ocpp:CP_TEST: send [2,"9a9ff96e-493e-4717-8e2f-9dfb341e6eef","Authorize",{ "idTag":"CARD_1234"}}
INFO:ocpp:CP_TEST: receive message [2,"9a9ff96e-493e-4717-8e2f-9dfb341e6eef","Authorize",{ "idTag":"CARD_1234"}}
INFO:MockSystemTest:[CSMS] <- Authorize: {'id_tag': 'CARD_1234'}
INFO:MockSystemTest:[CSMS] -> AuthorizeConf: Authorize(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"9a9ff96e-493e-4717-8e2f-9dfb341e6eef",{ "idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"9a9ff96e-493e-4717-8e2f-9dfb341e6eef",{ "idTagInfo":{"status":"Accepted"}}]
INFO:MockSystemTest:6.3 Authorize: Accepted
INFO:ocpp:CP_TEST: send [2,"b7b55e0f-76f7-41bb-b66c-e83751250da8","StartTransaction",{ "connectorId":1,"idTag":"CARD_1234","meterStart":0,"timestamp":"2026-01-05T15:26:59.799216"}}]
INFO:ocpp:CP_TEST: receive message [2,"b7b55e0f-76f7-41bb-b66c-e83751250da8","StartTransaction",{ "connectorId":1,"idTag":"CARD_1234","meterStart":0,"timestamp":"2026-01-05T15:26:59.799216"}}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'CARD_1234', 'meter_start': 0, 'timestamp': '2026-01-05T15:26:59.799216'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"b7b55e0f-76f7-41bb-b66c-e83751250da8",{ "transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"b7b55e0f-76f7-41bb-b66c-e83751250da8",{ "transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:6.4 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"2fcda2d5-747d-4631-9a88-6cb78a59361a","StatusNotification",{ "connectorId":1,"errorCode":"NoError","status":"Charging"}}
INFO:ocpp:CP_TEST: receive message [2,"2fcda2d5-747d-4631-9a88-6cb78a59361a","StatusNotification",{ "connectorId":1,"errorCode":"NoError","status":"Charging"}}
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"2fcda2d5-747d-4631-9a88-6cb78a59361a",{}]
INFO:ocpp:CP_TEST: receive message [3,"2fcda2d5-747d-4631-9a88-6cb78a59361a",{}]
INFO:MockSystemTest:6.5 StatusNotification: Charging
INFO:MockSystemTest:6.6 Waiting for MeterValues...
ERROR:MockSystemTest:FAIL: 6.6 No MeterValues received.
INFO:ocpp:CP_TEST: send [2,"7d83e49e-19c1-45e2-a28a-4b946a3e4b00","SetChargingProfile",{ "connectorId":1,"csChargingProfiles":{"chargingProfileId":1,"stackLevel":1,"chargingProfilePurpose":"TxProfile","chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":5000}]}}}]]
INFO:ocpp:CP_TEST: receive message [2,"7d83e49e-19c1-45e2-a28a-4b946a3e4b00","SetChargingProfile",{ "connectorId":1,"csChargingProfiles":{"chargingProfileId":1,"stackLevel":1,"chargingProfilePurpose":"TxProfile","chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":5000}]}}}]]
INFO:ocpp_16_interface:Received SetChargingProfile for connector 1
INFO:ocpp:CP_TEST: send [3,"7d83e49e-19c1-45e2-a28a-4b946a3e4b00",{ "status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"7d83e49e-19c1-45e2-a28a-4b946a3e4b00",{ "status":"Accepted"}}]
INFO:MockSystemTest:6.7 SetChargingProfile: Accepted
INFO:MockSystemTest:6.8 MeterValues received (assumed valid).
INFO:ocpp:CP_TEST: send [2,"c543b006-c4e8-4517-aa72-d31d5252c096","SetChargingProfile",{ "connectorId":1,"csChargingProfiles":{"chargingProfileId":1,"stackLevel":1,"chargingProfilePurpose":"TxProfile","chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":7000}]}}}]]
INFO:ocpp:CP_TEST: receive message [2,"c543b006-c4e8-4517-aa72-d31d5252c096","SetChargingProfile",{ "connectorId":1,"csChargingProfiles":{"chargingProfileId":1,"stackLevel":1,"chargingProfilePurpose":"TxProfile","chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":7000}]}}}]]
INFO:ocpp_16_interface:Received SetChargingProfile for connector 1
INFO:ocpp:CP_TEST: send [3,"c543b006-c4e8-4517-aa72-d31d5252c096",{ "status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"c543b006-c4e8-4517-aa72-d31d5252c096",{ "status":"Accepted"}}]
INFO:MockSystemTest:6.9 SetChargingProfile (Update): Accepted
INFO:MockSystemTest:6.10 MeterValues received.
INFO:ocpp:CP_TEST: send [2,"c68c7e0e-411a-4e59-a9b5-5f8df78fa2b3","StopTransaction",{ "meterStop":100,"timestamp":"2026-01-05T15:27:05.810181","transactionId":123,"reason":"Local"}}]
INFO:ocpp:CP_TEST: receive message [2,"c68c7e0e-411a-4e59-a9b5-5f8df78fa2b3","StopTransaction",{ "meterStop":100,"timestamp":"2026-01-05T15:27:05.810181","transactionId":123,"reason":"Local"}}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:05.810181', 'transaction_id': 123, 'reason': 'Local'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"c68c7e0e-411a-4e59-a9b5-5f8df78fa2b3",{ "idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"c68c7e0e-411a-4e59-a9b5-5f8df78fa2b3",{ "idTagInfo":{"status":"Accepted"}}]
```

```
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:6.11 StopTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"2bda2c85-6900-4fd4-9a4d-a084d03f8334","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:ocpp:CP_TEST: receive message [2,"2bda2c85-6900-4fd4-9a4d-a084d03f8334","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"2bda2c85-6900-4fd4-9a4d-a084d03f8334",{}]
INFO:ocpp:CP_TEST: receive message [3,"2bda2c85-6900-4fd4-9a4d-a084d03f8334",{}]
INFO:MockSystemTest:6.12 StatusNotification: Finishing
INFO:ocpp:CP_TEST: send [2,"53999f03-2cfc-4f1b-bdec-d768b94996d6","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"53999f03-2cfc-4f1b-bdec-d768b94996d6","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"53999f03-2cfc-4f1b-bdec-d768b94996d6",{}]
INFO:ocpp:CP_TEST: receive message [3,"53999f03-2cfc-4f1b-bdec-d768b94996d6",{}]
INFO:MockSystemTest:6.13 StatusNotification: Available
INFO:ocpp:CP_TEST: send [2,"d8b765f3-5fc8-4759-9615-4d3fa595f97e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"d8b765f3-5fc8-4759-9615-4d3fa595f97e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"d8b765f3-5fc8-4759-9615-4d3fa595f97e",{}]
INFO:ocpp:CP_TEST: receive message [3,"d8b765f3-5fc8-4759-9615-4d3fa595f97e",{}]
INFO:MockSystemTest:6.14 StatusNotification: Preparing
INFO:ocpp:CP_TEST: send [2,"4d4ac844-5055-423b-96a5-f230b1780766","RemoteStartTransaction",{"idTag":"REMOTE_TAG"}]
INFO:ocpp:CP_TEST: receive message [2,"4d4ac844-5055-423b-96a5-f230b1780766","RemoteStartTransaction",{"idTag":"REMOTE_TAG"}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: REMOTE_TAG
INFO:ocpp:CP_TEST: send [3,"4d4ac844-5055-423b-96a5-f230b1780766",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"4d1d9687-03bf-493f-8c6a-af312c799629","StartTransaction",{"connectorId":1,"idTag":"REMOTE_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:05.817432"}]
INFO:ocpp:CP_TEST: receive message [3,"4d4ac844-5055-423b-96a5-f230b1780766",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"4d1d9687-03bf-493f-8c6a-af312c799629","StartTransaction",{"connectorId":1,"idTag":"REMOTE_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:05.817432"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'REMOTE_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:05.817432'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:MockSystemTest:6.15 RemoteStartTransaction: Accepted
INFO:ocpp:CP_TEST: send [3,"4d1d9687-03bf-493f-8c6a-af312c799629",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"4d1d9687-03bf-493f-8c6a-af312c799629",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"bf66b304-a94b-44e9-b55a-5f42d3c587f2","StartTransaction",{"connectorId":1,"idTag":"REMOTE_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:05.818759"}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2,"bf66b304-a94b-44e9-b55a-5f42d3c587f2","StartTransaction",{"connectorId":1,"idTag":"REMOTE_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:05.818759"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'REMOTE_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:05.818759'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"bf66b304-a94b-44e9-b55a-5f42d3c587f2",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"bf66b304-a94b-44e9-b55a-5f42d3c587f2",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:6.16 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"a9091082-fd84-402c-9733-7de361d481ac","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"a9091082-fd84-402c-9733-7de361d481ac","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"a9091082-fd84-402c-9733-7de361d481ac",{}]
INFO:ocpp:CP_TEST: receive message [3,"a9091082-fd84-402c-9733-7de361d481ac",{}]
INFO:MockSystemTest:6.17 StatusNotification: Charging
INFO:MockSystemTest:6.18 MeterValues received.
INFO:ocpp:CP_TEST: send [2,"a4709fed-ecc4-4126-ba38-1019461e94e2","SetChargingProfile",{"connectorId":1,"csChargingProfiles":{"chargingProfileId":2,"stackLevel":1,"chargingProfilePurpose":"TxProfile","chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":7000}]}}}]]
INFO:ocpp:CP_TEST: receive message [2,"a4709fed-ecc4-4126-ba38-1019461e94e2","SetChargingProfile",{"connectorId":1,"csChargingProfiles":{"chargingProfileId":2,"stackLevel":1,"chargingProfilePurpose":"TxProfile","chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":7000}]}}}]]
INFO:ocpp_16_interface:Received SetChargingProfile for connector 1
```

```
INFO:ocpp:CP_TEST: send [3,"a4709fed-ecc4-4126-ba38-1019461e94e2",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"a4709fed-ecc4-4126-ba38-1019461e94e2",{"status":"Accepted"}]
INFO:MockSystemTest:6.19 SetChargingProfile: Accepted
INFO:MockSystemTest:6.20 MeterValues received.
INFO:ocpp:CP_TEST: send [2,"84b5bd11-bfe3-4c08-ac16-52b4cfc6a309","SetChargingProfile",{"connectorId":1,"csChargingProfiles":{"chargingProfileId":2,"stackLevel":1,"chargingProfilePurpose":"TxProfile",
chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":3000}]}}}]
INFO:ocpp:CP_TEST: receive message [2,"84b5bd11-bfe3-4c08-ac16-52b4cfc6a309","SetChargingProfile",{"connectorId":1,"csChargingProfiles":{"chargingProfileId":2,"stackLevel":1,"chargingProfilePurpose":"TxProfile",
,"chargingProfileKind":"Absolute","chargingSchedule":{"chargingRateUnit":"W","chargingSchedulePeriod":[{"startPeriod":0,"limit":3000}]}}}]
INFO:ocpp_16_interface:Received SetChargingProfile for connector 1
INFO:ocpp:CP_TEST: send [3,"84b5bd11-bfe3-4c08-ac16-52b4cfc6a309",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"84b5bd11-bfe3-4c08-ac16-52b4cfc6a309",{"status":"Accepted"}]
INFO:MockSystemTest:6.21 SetChargingProfile: Accepted
INFO:MockSystemTest:6.22 MeterValues received.
INFO:ocpp:CP_TEST: send [2,"197e9c3b-43b8-4bab-8ff5-85f62c6df649","RemoteStopTransaction",{"transactionId":123}]
INFO:ocpp:CP_TEST: receive message [2,"197e9c3b-43b8-4bab-8ff5-85f62c6df649","RemoteStopTransaction",{"transactionId":123}]
INFO:ocpp_16_interface:Received RemoteStopTransaction for transaction_id: 123
INFO:ocpp:CP_TEST: send [3,"197e9c3b-43b8-4bab-8ff5-85f62c6df649",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"197e9c3b-43b8-4bab-8ff5-85f62c6df649",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"b5bce943-b3c1-445b-8f91-60595fcff687","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:11.830985","transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:6.23 RemoteStopTransaction: Accepted
INFO:ocpp:CP_TEST: receive message [2,"b5bce943-b3c1-445b-8f91-60595fcff687","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:11.830985","transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:11.830985', 'transaction_id': 123, 'reason': 'Remote'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"b5bce943-b3c1-445b-8f91-60595fcff687",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"b5bce943-b3c1-445b-8f91-60595fcff687",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:6.24 StopTransaction: Triggered by RemoteStop
INFO:ocpp:CP_TEST: send [2,"7c9ed217-4ece-4685-bfbb-61cc8e7f8ebf",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}}]
INFO:ocpp:CP_TEST: receive message [2,"7c9ed217-4ece-4685-bfbb-61cc8e7f8ebf",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"7c9ed217-4ece-4685-bfbb-61cc8e7f8ebf",{}]
INFO:ocpp:CP_TEST: receive message [3,"7c9ed217-4ece-4685-bfbb-61cc8e7f8ebf",{}]
INFO:MockSystemTest:6.25 StatusNotification: Finishing
INFO:ocpp:CP_TEST: send [2,"695f56f2-7b5e-472f-88ec-650867190ddf",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"695f56f2-7b5e-472f-88ec-650867190ddf",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"695f56f2-7b5e-472f-88ec-650867190ddf",{}]
INFO:ocpp:CP_TEST: receive message [3,"695f56f2-7b5e-472f-88ec-650867190ddf",{}]
INFO:MockSystemTest:6.26 StatusNotification: Available
INFO:MockSystemTest:
--- 7. Abnormal Operation Verification (Detailed) ---
INFO:MockSystemTest:--- 7.1 Remote Start (Unplugged) ---
INFO:ocpp:CP_TEST: send [2,"e76d3bd0-06b2-455e-a879-60f56c50be2b",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"e76d3bd0-06b2-455e-a879-60f56c50be2b",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"e76d3bd0-06b2-455e-a879-60f56c50be2b",{}]
INFO:ocpp:CP_TEST: receive message [3,"e76d3bd0-06b2-455e-a879-60f56c50be2b",{}]
INFO:MockSystemTest:7.1.1 Status: Available
INFO:ocpp:CP_TEST: send [2,"6037fa7e-4beb-4a1e-b5c2-5d70f18daa69",{"RemoteStartTransaction",{"idTag":"REMOTE_UNPLUGGED"}}]
INFO:ocpp:CP_TEST: receive message [2,"6037fa7e-4beb-4a1e-b5c2-5d70f18daa69",{"RemoteStartTransaction",{"idTag":"REMOTE_UNPLUGGED"}}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: REMOTE_UNPLUGGED
INFO:ocpp:CP_TEST: send [3,"6037fa7e-4beb-4a1e-b5c2-5d70f18daa69",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"e7f068ee-9506-4800-97e5-d1ce96d88ba6",{"StartTransaction",{"connectorId":1,"idTag":"REMOTE_UNPLUGGED","meterStart":0,"timestamp":"2026-01-05T15:27:12.840707"}}]
INFO:ocpp:CP_TEST: receive message [3,"6037fa7e-4beb-4a1e-b5c2-5d70f18daa69",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"e7f068ee-9506-4800-97e5-d1ce96d88ba6",{"StartTransaction",{"connectorId":1,"idTag":"REMOTE_UNPLUGGED","meterStart":0,"timestamp":"2026-01-05T15:27:12.840707"}}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'REMOTE_UNPLUGGED', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:12.840707'}
```



```
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:MockSystemTest:7.1.2 RemoteStart (Unplugged): Accepted
INFO:MockSystemTest:--- 7.2 Concurrent Remote Start ---
INFO:ocpp:CP_TEST: send [3,"e7f068ee-9506-4800-97e5-d1ce96d88ba6",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"e7f068ee-9506-4800-97e5-d1ce96d88ba6",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"6a32027d-4b4d-4355-a88c-03a939cc8da5",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Preparing"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2,"6a32027d-4b4d-4355-a88c-03a939cc8da5",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Preparing"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"6a32027d-4b4d-4355-a88c-03a939cc8da5",{}]
INFO:ocpp:CP_TEST: receive message [3,"6a32027d-4b4d-4355-a88c-03a939cc8da5",{}]
INFO:MockSystemTest:7.2.1 Status: Preparing
INFO:MockSystemTest:--- 7.2 Remote Start Same Connector ID ---
INFO:ocpp:CP_TEST: send [2,"bb379f87-37e1-4e91-a8d5-ced46214a4b3",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"bb379f87-37e1-4e91-a8d5-ced46214a4b3",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"bb379f87-37e1-4e91-a8d5-ced46214a4b3",{}]
INFO:ocpp:CP_TEST: receive message [3,"bb379f87-37e1-4e91-a8d5-ced46214a4b3",{}]
INFO:MockSystemTest:7.2.1 Status: Unplugged
INFO:ocpp:CP_TEST: send [2,"7d3cfdb8-8ba6-43c7-8c4e-83cd20930338",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Preparing"}}]
INFO:ocpp:CP_TEST: receive message [2,"7d3cfdb8-8ba6-43c7-8c4e-83cd20930338",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Preparing"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"7d3cfdb8-8ba6-43c7-8c4e-83cd20930338",{}]
INFO:ocpp:CP_TEST: receive message [3,"7d3cfdb8-8ba6-43c7-8c4e-83cd20930338",{}]
INFO:MockSystemTest:7.2.2 Status: Plugged
INFO:ocpp:CP_TEST: send [2,"f8a2256e-9691-4962-85ed-7208f5db4caf",{"RemoteStartTransaction":{"idTag":"REMOTE_1","connectorId":1}}]
INFO:ocpp:CP_TEST: receive message [2,"f8a2256e-9691-4962-85ed-7208f5db4caf",{"RemoteStartTransaction":{"idTag":"REMOTE_1","connectorId":1}}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: REMOTE_1
INFO:ocpp:CP_TEST: send [3,"f8a2256e-9691-4962-85ed-7208f5db4caf",{"status":"Rejected"}]
INFO:ocpp:CP_TEST: receive message [3,"f8a2256e-9691-4962-85ed-7208f5db4caf",{"status":"Rejected"}]
INFO:MockSystemTest:7.2.3 RemoteStart (1): Rejected
INFO:ocpp:CP_TEST: send [2,"81195fb8-0802-4d80-9ca4-d542fdc8ff1e",{"RemoteStartTransaction":{"idTag":"REMOTE_2","connectorId":1}}]
INFO:ocpp:CP_TEST: receive message [2,"81195fb8-0802-4d80-9ca4-d542fdc8ff1e",{"RemoteStartTransaction":{"idTag":"REMOTE_2","connectorId":1}}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: REMOTE_2
INFO:ocpp:CP_TEST: send [3,"81195fb8-0802-4d80-9ca4-d542fdc8ff1e",{"status":"Rejected"}]
INFO:ocpp:CP_TEST: receive message [3,"81195fb8-0802-4d80-9ca4-d542fdc8ff1e",{"status":"Rejected"}]
INFO:MockSystemTest:7.2.4 RemoteStart (2) Same ID: Rejected
WARNING:ocpp_16_interface:Transaction already in progress for connector 1.
INFO:MockSystemTest:7.2.5 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"1d216e5d-8f3b-46c3-81d3-25902b8de6c8",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Charging"}}]
INFO:ocpp:CP_TEST: receive message [2,"1d216e5d-8f3b-46c3-81d3-25902b8de6c8",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Charging"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"1d216e5d-8f3b-46c3-81d3-25902b8de6c8",{}]
INFO:ocpp:CP_TEST: receive message [3,"1d216e5d-8f3b-46c3-81d3-25902b8de6c8",{}]
INFO:MockSystemTest:7.2.6 Status: Charging
INFO:MockSystemTest:7.2.7 MeterValues: Simulated
INFO:ocpp:CP_TEST: send [2,"a489aa29-e7b5-4ee7-bf5f-4416610fa5ed",{"RemoteStopTransaction":{"transactionId":123}}]
INFO:ocpp:CP_TEST: receive message [2,"a489aa29-e7b5-4ee7-bf5f-4416610fa5ed",{"RemoteStopTransaction":{"transactionId":123}}]
INFO:ocpp_16_interface:Received RemoteStopTransaction for transaction_id: 123
INFO:ocpp:CP_TEST: send [3,"a489aa29-e7b5-4ee7-bf5f-4416610fa5ed",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"a489aa29-e7b5-4ee7-bf5f-4416610fa5ed",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"39faca05-5356-40d1-a391-21d5a1094708",{"StopTransaction":{"meterStop":100,"timestamp":"2026-01-05T15:27:13.854353","transactionId":123,"reason":"Remote"}}]
INFO:MockSystemTest:7.2.8 RemoteStop: Sent
INFO:ocpp:CP_TEST: receive message [2,"39faca05-5356-40d1-a391-21d5a1094708",{"StopTransaction":{"meterStop":100,"timestamp":"2026-01-05T15:27:13.854353","transactionId":123,"reason":"Remote"}}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:13.854353', 'transaction_id': 123, 'reason': 'Remote'}
```

```
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"39faca05-5356-40d1-a391-21d5a1094708",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"39faca05-5356-40d1-a391-21d5a1094708",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"4d6194a7-feae-4aee-860b-4d45725f1f35","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:13.855446","transactionId":123,"reason":"Remote"}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:ocpp:CP_TEST: receive message [2,"4d6194a7-feae-4aee-860b-4d45725f1f35","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:13.855446","transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:13.855446', 'transaction_id': 123, 'reason': 'Remote'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"4d6194a7-feae-4aee-860b-4d45725f1f35",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"4d6194a7-feae-4aee-860b-4d45725f1f35",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:7.2.9 StopTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"6dc1285f-d8d7-48e6-96cf-87648cad7ef7","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:ocpp:CP_TEST: receive message [2,"6dc1285f-d8d7-48e6-96cf-87648cad7ef7","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"6dc1285f-d8d7-48e6-96cf-87648cad7ef7",{}]
INFO:ocpp:CP_TEST: receive message [3,"6dc1285f-d8d7-48e6-96cf-87648cad7ef7",{}]
INFO:ocpp:CP_TEST: send [2,"0fb3188d-d8ce-472a-b1e4-a291f81404b4","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"0fb3188d-d8ce-472a-b1e4-a291f81404b4","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"0fb3188d-d8ce-472a-b1e4-a291f81404b4",{}]
INFO:ocpp:CP_TEST: receive message [3,"0fb3188d-d8ce-472a-b1e4-a291f81404b4",{}]
INFO:MockSystemTest:7.2.11 Status: Unplugged
INFO:MockSystemTest:--- 7.3 Swap Card ---
INFO:ocpp:CP_TEST: send [2,"24892ce2-9479-431c-874e-92f2915f10eb","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"24892ce2-9479-431c-874e-92f2915f10eb","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"24892ce2-9479-431c-874e-92f2915f10eb",{}]
INFO:ocpp:CP_TEST: receive message [3,"24892ce2-9479-431c-874e-92f2915f10eb",{}]
INFO:ocpp:CP_TEST: send [2,"3b2c8e6a-a8a5-44cb-b06b-eab3d190e1a7","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"3b2c8e6a-a8a5-44cb-b06b-eab3d190e1a7","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"3b2c8e6a-a8a5-44cb-b06b-eab3d190e1a7",{}]
INFO:ocpp:CP_TEST: receive message [3,"3b2c8e6a-a8a5-44cb-b06b-eab3d190e1a7",{}]
INFO:ocpp:CP_TEST: send [2,"815a8005-b1a6-43ad-8d6b-aa79ac619d6c","Authorize",{"idTag":"INVALID_CARD"}]
INFO:ocpp:CP_TEST: receive message [2,"815a8005-b1a6-43ad-8d6b-aa79ac619d6c","Authorize",{"idTag":"INVALID_CARD"}]
INFO:MockSystemTest:[CSMS] <- Authorize: {'id_tag': 'INVALID_CARD'}
INFO:MockSystemTest:[CSMS] -> AuthorizeConf: Authorize(id_tag_info={'status': <AuthorizationStatus.invalid: 'Invalid'>})
INFO:ocpp:CP_TEST: send [3,"815a8005-b1a6-43ad-8d6b-aa79ac619d6c",{"idTagInfo":{"status":"Invalid"}}]
INFO:ocpp:CP_TEST: receive message [3,"815a8005-b1a6-43ad-8d6b-aa79ac619d6c",{"idTagInfo":{"status":"Invalid"}}]
INFO:MockSystemTest:7.3.3 Authorize (Invalid): Invalid
INFO:ocpp:CP_TEST: send [2,"cdfaf186-5b9e-4c5a-9065-f74d6336029c","Authorize",{"idTag":"VALID_CARD"}]
INFO:ocpp:CP_TEST: receive message [2,"cdfaf186-5b9e-4c5a-9065-f74d6336029c","Authorize",{"idTag":"VALID_CARD"}]
INFO:MockSystemTest:[CSMS] <- Authorize: {'id_tag': 'VALID_CARD'}
INFO:MockSystemTest:[CSMS] -> AuthorizeConf: Authorize(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"cdfaf186-5b9e-4c5a-9065-f74d6336029c",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"cdfaf186-5b9e-4c5a-9065-f74d6336029c",{"idTagInfo":{"status":"Accepted"}}]
INFO:MockSystemTest:7.3.4 Authorize (Accepted): Accepted
INFO:ocpp:CP_TEST: send [2,"53d393b0-bbae-419d-adf8-2857b9e1f898","StartTransaction",{"connectorId":1,"idTag":"VALID_CARD","meterStart":0,"timestamp":"2026-01-05T15:27:13.867990"}]
INFO:ocpp:CP_TEST: receive message [2,"53d393b0-bbae-419d-adf8-2857b9e1f898","StartTransaction",{"connectorId":1,"idTag":"VALID_CARD","meterStart":0,"timestamp":"2026-01-05T15:27:13.867990"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'VALID_CARD', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:13.867990'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"53d393b0-bbae-419d-adf8-2857b9e1f898",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"53d393b0-bbae-419d-adf8-2857b9e1f898",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
```

```
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:7.3.5 StartTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"5fa48e59-260b-4d3b-aff3-7f704efe178e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"5fa48e59-260b-4d3b-aff3-7f704efe178e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"5fa48e59-260b-4d3b-aff3-7f704efe178e",{}]
INFO:ocpp:CP_TEST: receive message [3,"5fa48e59-260b-4d3b-aff3-7f704efe178e",{}]
INFO:ocpp:CP_TEST: send [2,"e3215b36-958d-4342-8512-bbd36dda90f8","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:14.871532","transactionId":123,"reason":"Local"}]
INFO:ocpp:CP_TEST: receive message [2,"e3215b36-958d-4342-8512-bbd36dda90f8","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:14.871532","transactionId":123,"reason":"Local"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:14.871532', 'transaction_id': 123, 'reason': 'Local'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"e3215b36-958d-4342-8512-bbd36dda90f8",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"e3215b36-958d-4342-8512-bbd36dda90f8",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:7.3.9 StopTransaction: Sent
INFO:ocpp:CP_TEST: send [2,"8a00d023-d19b-44c9-9a59-d0c9e84aa162","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:ocpp:CP_TEST: receive message [2,"8a00d023-d19b-44c9-9a59-d0c9e84aa162","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Finishing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"8a00d023-d19b-44c9-9a59-d0c9e84aa162",{}]
INFO:ocpp:CP_TEST: receive message [3,"8a00d023-d19b-44c9-9a59-d0c9e84aa162",{}]
INFO:ocpp:CP_TEST: send [2,"54179e3a-2086-4923-81a1-a5cb9cd75921","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"54179e3a-2086-4923-81a1-a5cb9cd75921","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"54179e3a-2086-4923-81a1-a5cb9cd75921",{}]
INFO:ocpp:CP_TEST: receive message [3,"54179e3a-2086-4923-81a1-a5cb9cd75921",{}]
INFO:MockSystemTest:--- 7.4 Emergency Stop ---
INFO:ocpp:CP_TEST: send [2,"2f2bf36e-048e-4553-9712-e0a71f9fcdbe","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"2f2bf36e-048e-4553-9712-e0a71f9fcdbe","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"2f2bf36e-048e-4553-9712-e0a71f9fcdbe",{}]
INFO:ocpp:CP_TEST: receive message [3,"2f2bf36e-048e-4553-9712-e0a71f9fcdbe",{}]
INFO:ocpp:CP_TEST: send [2,"ce23ccdb-708b-4170-afaf-db9fae1a3134","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"ce23ccdb-708b-4170-afaf-db9fae1a3134","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"ce23ccdb-708b-4170-afaf-db9fae1a3134",{}]
INFO:ocpp:CP_TEST: receive message [3,"ce23ccdb-708b-4170-afaf-db9fae1a3134",{}]
INFO:ocpp:CP_TEST: send [2,"afcd3fbb-f5d8-4eb2-bfc9-105b2dfa21c7","RemoteStartTransaction",{"idTag":"EMERGENCY_TAG"}]
INFO:ocpp:CP_TEST: receive message [2,"afcd3fbb-f5d8-4eb2-bfc9-105b2dfa21c7","RemoteStartTransaction",{"idTag":"EMERGENCY_TAG"}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: EMERGENCY_TAG
INFO:ocpp:CP_TEST: send [3,"afcd3fbb-f5d8-4eb2-bfc9-105b2dfa21c7",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"9d4bac9c-c3a8-43e2-aa66-1eda5b05af25","StartTransaction",{"connectorId":1,"idTag":"EMERGENCY_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:14.882026"}]
INFO:ocpp:CP_TEST: receive message [3,"afcd3fbb-f5d8-4eb2-bfc9-105b2dfa21c7",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"9d4bac9c-c3a8-43e2-aa66-1eda5b05af25","StartTransaction",{"connectorId":1,"idTag":"EMERGENCY_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:14.882026"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'EMERGENCY_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:14.882026'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"9d4bac9c-c3a8-43e2-aa66-1eda5b05af25",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"9d4bac9c-c3a8-43e2-aa66-1eda5b05af25",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"b9767d1b-0c29-451d-a485-608b7d243159","StartTransaction",{"connectorId":1,"idTag":"EMERGENCY_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:14.883347"}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2,"b9767d1b-0c29-451d-a485-608b7d243159","StartTransaction",{"connectorId":1,"idTag":"EMERGENCY_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:14.883347"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'EMERGENCY_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:14.883347'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
```

```
INFO:ocpp:CP_TEST: send [3,"b9767d1b-0c29-451d-a485-608b7d243159",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"b9767d1b-0c29-451d-a485-608b7d243159",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: send [2,"80915237-6439-43ba-911f-3dab14d34671",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"80915237-6439-43ba-911f-3dab14d34671",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"80915237-6439-43ba-911f-3dab14d34671",{}]
INFO:ocpp:CP_TEST: receive message [3,"80915237-6439-43ba-911f-3dab14d34671",{}]
INFO:ocpp:CP_TEST: send [2,"fb984404-07b9-41e0-ae07-2245bfe33b59",{"StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:15.887396","transactionId":123,"reason":"EmergencyStop"}}]
INFO:ocpp:CP_TEST: receive message [2,"fb984404-07b9-41e0-ae07-2245bfe33b59",{"StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:15.887396","transactionId":123,"reason":"EmergencyStop"}}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:15.887396', 'transaction_id': 123, 'reason': 'EmergencyStop'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"fb984404-07b9-41e0-ae07-2245bfe33b59",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"fb984404-07b9-41e0-ae07-2245bfe33b59",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:7.4.7 StopTransaction (Emergency): Sent
INFO:ocpp:CP_TEST: send [2,"b3363138-7c87-4787-a445-cac5ad78ab0a",{"StatusNotification",{"connectorId":1,"errorCode":"OtherError","status":"Faulted","info":"EmergencyStop"}}]
INFO:ocpp:CP_TEST: receive message [2,"b3363138-7c87-4787-a445-cac5ad78ab0a",{"StatusNotification",{"connectorId":1,"errorCode":"OtherError","status":"Faulted","info":"EmergencyStop"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'OtherError', 'status': 'Faulted', 'info': 'EmergencyStop'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"b3363138-7c87-4787-a445-cac5ad78ab0a",{}]
INFO:ocpp:CP_TEST: receive message [3,"b3363138-7c87-4787-a445-cac5ad78ab0a",{}]
INFO:MockSystemTest:7.4.8 Status: Faulted
INFO:ocpp:CP_TEST: send [2,"c5d9c862-2509-4827-9548-2a8985b4b11e",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"c5d9c862-2509-4827-9548-2a8985b4b11e",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"c5d9c862-2509-4827-9548-2a8985b4b11e",{}]
INFO:ocpp:CP_TEST: receive message [3,"c5d9c862-2509-4827-9548-2a8985b4b11e",{}]
INFO:MockSystemTest:7.4.10 Status: Available
INFO:MockSystemTest:--- 7.5 Open Door ---
INFO:ocpp:CP_TEST: send [2,"c8c27bd4-a179-4a8f-8a0a-7b5dd1e53ab1",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"c8c27bd4-a179-4a8f-8a0a-7b5dd1e53ab1",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"c8c27bd4-a179-4a8f-8a0a-7b5dd1e53ab1",{}]
INFO:ocpp:CP_TEST: receive message [3,"c8c27bd4-a179-4a8f-8a0a-7b5dd1e53ab1",{}]
INFO:ocpp:CP_TEST: send [2,"60d79bf0-4239-4eb3-beb0-893cd1121014",{"StatusNotification",{"connectorId":1,"errorCode":"OtherError","status":"Faulted","info":"DoorOpen"}}]
INFO:ocpp:CP_TEST: receive message [2,"60d79bf0-4239-4eb3-beb0-893cd1121014",{"StatusNotification",{"connectorId":1,"errorCode":"OtherError","status":"Faulted","info":"DoorOpen"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'OtherError', 'status': 'Faulted', 'info': 'DoorOpen'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"60d79bf0-4239-4eb3-beb0-893cd1121014",{}]
INFO:ocpp:CP_TEST: receive message [3,"60d79bf0-4239-4eb3-beb0-893cd1121014",{}]
INFO:MockSystemTest:7.5.2 Status: DoorOpen
INFO:ocpp:CP_TEST: send [2,"94eddfb8-c661-4f71-ab8c-3b3ec9140f8a",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"94eddfb8-c661-4f71-ab8c-3b3ec9140f8a",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"94eddfb8-c661-4f71-ab8c-3b3ec9140f8a",{}]
INFO:ocpp:CP_TEST: receive message [3,"94eddfb8-c661-4f71-ab8c-3b3ec9140f8a",{}]
INFO:MockSystemTest:7.5.3 Status: Closed/Available
INFO:MockSystemTest:--- 7.6 Power Loss (Single) ---
INFO:ocpp:CP_TEST: send [2,"0c29e06b-0359-4eee-adcb-c8b113df9266",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"0c29e06b-0359-4eee-adcb-c8b113df9266",{"StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"0c29e06b-0359-4eee-adcb-c8b113df9266",{}]
```

```
INFO:ocpp:CP_TEST: receive message [3,"0c29e06b-0359-4eee-adcb-c8b113df9266",{}]  
INFO:ocpp:CP_TEST: send [2,"8b82eeb3-f2da-440d-8f61-0b948afc895e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]  
INFO:ocpp:CP_TEST: receive message [2,"8b82eeb3-f2da-440d-8f61-0b948afc895e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]  
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}  
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}  
INFO:ocpp:CP_TEST: send [3,"8b82eeb3-f2da-440d-8f61-0b948afc895e",{}]  
INFO:ocpp:CP_TEST: receive message [3,"8b82eeb3-f2da-440d-8f61-0b948afc895e",{}]  
INFO:ocpp:CP_TEST: send [2,"f3531393-6eef-4ef9-93a0-6fc2a07da71e","RemoteStartTransaction",{"idTag":"POWER_TAG"}]  
INFO:ocpp:CP_TEST: receive message [2,"f3531393-6eef-4ef9-93a0-6fc2a07da71e","RemoteStartTransaction",{"idTag":"POWER_TAG"}]  
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: POWER_TAG  
INFO:ocpp:CP_TEST: send [3,"f3531393-6eef-4ef9-93a0-6fc2a07da71e",{"status":"Accepted"}]  
INFO:ocpp:CP_TEST: send [2,"ff605151-964e-434f-93a4-89f2db0d5d87","StartTransaction",{"connectorId":1,"idTag":"POWER_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:15.901994"}]  
INFO:ocpp:CP_TEST: receive message [3,"f3531393-6eef-4ef9-93a0-6fc2a07da71e",{"status":"Accepted"}]  
INFO:ocpp:CP_TEST: receive message [2,"ff605151-964e-434f-93a4-89f2db0d5d87","StartTransaction",{"connectorId":1,"idTag":"POWER_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:15.901994"}]  
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'POWER_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:15.901994'}  
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})  
INFO:ocpp:CP_TEST: send [3,"ff605151-964e-434f-93a4-89f2db0d5d87",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp:CP_TEST: receive message [3,"ff605151-964e-434f-93a4-89f2db0d5d87",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp:CP_TEST: send [2,"dfcb3b38-313f-4e60-8f57-2ee7082e5455","StartTransaction",{"connectorId":1,"idTag":"POWER_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:15.903208"}]  
INFO:ocpp_16_interface:Transaction started on 1: 123  
INFO:MockSystemTest:HARDWARE: Charger started.  
INFO:ocpp:CP_TEST: receive message [2,"dfcb3b38-313f-4e60-8f57-2ee7082e5455","StartTransaction",{"connectorId":1,"idTag":"POWER_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:15.903208"}]  
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'POWER_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:15.903208'}  
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})  
INFO:ocpp:CP_TEST: send [3,"dfcb3b38-313f-4e60-8f57-2ee7082e5455",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp:CP_TEST: receive message [3,"dfcb3b38-313f-4e60-8f57-2ee7082e5455",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp_16_interface:Transaction started on 1: 123  
INFO:MockSystemTest:HARDWARE: Charger started.  
INFO:ocpp:CP_TEST: send [2,"e08f1c65-d7ad-4fe7-9aa7-5e85314e331e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]  
INFO:ocpp:CP_TEST: receive message [2,"e08f1c65-d7ad-4fe7-9aa7-5e85314e331e","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]  
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}  
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}  
INFO:ocpp:CP_TEST: send [3,"e08f1c65-d7ad-4fe7-9aa7-5e85314e331e",{}]  
INFO:ocpp:CP_TEST: receive message [3,"e08f1c65-d7ad-4fe7-9aa7-5e85314e331e",{}]  
INFO:ocpp:CP_TEST: send [2,"b8dcffe7-ba95-4343-bf98-626f910e4d96","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:16.907780","transactionId":123,"reason":"PowerLoss"}]  
INFO:ocpp:CP_TEST: receive message [2,"b8dcffe7-ba95-4343-bf98-626f910e4d96","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:16.907780","transactionId":123,"reason":"PowerLoss"}]  
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:16.907780', 'transaction_id': 123, 'reason': 'PowerLoss'}  
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})  
INFO:ocpp:CP_TEST: send [3,"b8dcffe7-ba95-4343-bf98-626f910e4d96",{"idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp:CP_TEST: receive message [3,"b8dcffe7-ba95-4343-bf98-626f910e4d96",{"idTagInfo":{"status":"Accepted"}}]  
INFO:ocpp_16_interface:Transaction stopped on 1: 123  
INFO:MockSystemTest:HARDWARE: Charger stopped.  
INFO:MockSystemTest:7.6.7 StopTransaction (PowerLoss): Sent  
INFO:ocpp:CP_TEST: send [2,"dee7fcf2-7f41-41c6-a9e9-fe2077358424","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]  
INFO:ocpp:CP_TEST: receive message [2,"dee7fcf2-7f41-41c6-a9e9-fe2077358424","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]  
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}  
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:27:16.911873', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)  
INFO:ocpp:CP_TEST: send [3,"dee7fcf2-7f41-41c6-a9e9-fe2077358424",{"currentTime":"2026-01-05T15:27:16.911873","interval":10,"status":"Accepted"}]  
INFO:ocpp:CP_TEST: receive message [3,"dee7fcf2-7f41-41c6-a9e9-fe2077358424",{"currentTime":"2026-01-05T15:27:16.911873","interval":10,"status":"Accepted"}]  
INFO:ocpp_16_interface:Connected to central system.  
INFO:MockSystemTest:HARDWARE: Charger started.  
INFO:MockSystemTest:7.6.8 BootNotification: Sent  
INFO:ocpp:CP_TEST: send [2,"a12cc9d6-206a-4373-a024-0aa5946e04dc","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]  
INFO:ocpp:CP_TEST: receive message [2,"a12cc9d6-206a-4373-a024-0aa5946e04dc","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]  
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}  
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}  
INFO:ocpp:CP_TEST: send [3,"a12cc9d6-206a-4373-a024-0aa5946e04dc",{}]  
INFO:ocpp:CP_TEST: receive message [3,"a12cc9d6-206a-4373-a024-0aa5946e04dc",{}]  
INFO:ocpp:CP_TEST: send [2,"6a338992-45a2-4cfa-8ade-330d1e0297bb","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]  
INFO:ocpp:CP_TEST: receive message [2,"6a338992-45a2-4cfa-8ade-330d1e0297bb","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
```

```
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"6a338992-45a2-4cfa-8ade-330d1e0297bb",{}]
INFO:ocpp:CP_TEST: receive message [3,"6a338992-45a2-4cfa-8ade-330d1e0297bb",{}]
INFO:MockSystemTest:--- 7.7 Local List Offline ---
INFO:ocpp:CP_TEST: send [2,"bd1dc4dd-bbd2-4595-85aa-d0b363f86f9b","ChangeConfiguration",{"key":"LocalAuthorizeOffline","value":"True"}]
INFO:ocpp:CP_TEST: receive message [2,"bd1dc4dd-bbd2-4595-85aa-d0b363f86f9b","ChangeConfiguration",{"key":"LocalAuthorizeOffline","value":"True"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for LocalAuthorizeOffline to True
INFO:ocpp:CP_TEST: send [3,"bd1dc4dd-bbd2-4595-85aa-d0b363f86f9b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"bd1dc4dd-bbd2-4595-85aa-d0b363f86f9b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"2a486614-a054-43b9-9c82-d6e40f55445e","SendLocalList",{"listVersion":1,"updateType":"Full","localAuthorizationList":[{"idTag":"OFFLINE_TAG","idTagInfo":{"status":"Accepted"}}]}]
INFO:ocpp:CP_TEST: receive message [2,"2a486614-a054-43b9-9c82-d6e40f55445e","SendLocalList",{"listVersion":1,"updateType":"Full","localAuthorizationList":[{"idTag":"OFFLINE_TAG","idTagInfo":{"status":"Accepted"}}]}]]]
INFO:ocpp_16_interface:Received SendLocalList
INFO:ocpp:CP_TEST: send [3,"2a486614-a054-43b9-9c82-d6e40f55445e",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"2a486614-a054-43b9-9c82-d6e40f55445e",{"status":"Accepted"}]
INFO:MockSystemTest:7.7.2 Start by Local List (Offline Mock)
INFO:ocpp:CP_TEST: send [2,"1fef4fa4-82f6-405d-8d4f-f76ff653adc2","StartTransaction",{"connectorId":1,"idTag":"OFFLINE_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:16.919449"}]
INFO:ocpp:CP_TEST: receive message [2,"1fef4fa4-82f6-405d-8d4f-f76ff653adc2","StartTransaction",{"connectorId":1,"idTag":"OFFLINE_TAG","meterStart":0,"timestamp":"2026-01-05T15:27:16.919449"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'OFFLINE_TAG', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:16.919449'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': 'Accepted'})
INFO:ocpp:CP_TEST: send [3,"1fef4fa4-82f6-405d-8d4f-f76ff653adc2",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"1fef4fa4-82f6-405d-8d4f-f76ff653adc2",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:7.7.2 StartTransaction: Success
INFO:ocpp:CP_TEST: send [2,"f16c689b-a385-4184-9e0e-0f5af7eb156d","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:16.921379","transactionId":123,"reason":"Local"}]
INFO:ocpp:CP_TEST: receive message [2,"f16c689b-a385-4184-9e0e-0f5af7eb156d","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:16.921379","transactionId":123,"reason":"Local"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:16.921379', 'transaction_id': 123, 'reason': 'Local'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': 'Accepted'})
INFO:ocpp:CP_TEST: send [3,"f16c689b-a385-4184-9e0e-0f5af7eb156d",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"f16c689b-a385-4184-9e0e-0f5af7eb156d",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:7.7.3 StopTransaction: Success
INFO:MockSystemTest:
--- 8. Dual Connector Verification (Detailed) ---
INFO:MockSystemTest:--- 8.1 Concurrent Remote Start ---
INFO:ocpp:CP_TEST: send [2,"71119669-7b68-4aff-80f2-4a222014e870","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"71119669-7b68-4aff-80f2-4a222014e870","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"71119669-7b68-4aff-80f2-4a222014e870",{}]
INFO:ocpp:CP_TEST: receive message [3,"71119669-7b68-4aff-80f2-4a222014e870",{}]
INFO:ocpp:CP_TEST: send [2,"06bb2e25-048f-4450-9c21-4843d85b4876","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"06bb2e25-048f-4450-9c21-4843d85b4876","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"06bb2e25-048f-4450-9c21-4843d85b4876",{}]
INFO:ocpp:CP_TEST: receive message [3,"06bb2e25-048f-4450-9c21-4843d85b4876",{}]
INFO:MockSystemTest:8.1.1 Status (Both): Available
INFO:ocpp:CP_TEST: send [2,"296e4f4f-06da-4c08-b981-b8366ea9552f","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"296e4f4f-06da-4c08-b981-b8366ea9552f","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"296e4f4f-06da-4c08-b981-b8366ea9552f",{}]
INFO:ocpp:CP_TEST: receive message [3,"296e4f4f-06da-4c08-b981-b8366ea9552f",{}]
INFO:ocpp:CP_TEST: send [2,"8e422b13-69ed-4101-8a0c-c062b9213d27","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"8e422b13-69ed-4101-8a0c-c062b9213d27","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Preparing'}
```

```
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"8e422b13-69ed-4101-8a0c-c062b9213d27",{}]
INFO:ocpp:CP_TEST: receive message [3,"8e422b13-69ed-4101-8a0c-c062b9213d27",{}]
INFO:MockSystemTest:8.1.2 Plug Both: Preparing
INFO:ocpp:CP_TEST: send [2,"e5125a08-f864-4524-89f8-553942875273","RemoteStartTransaction",{"idTag":"DUAL_1","connectorId":1}]
INFO:ocpp:CP_TEST: receive message [2,"e5125a08-f864-4524-89f8-553942875273","RemoteStartTransaction",{"idTag":"DUAL_1","connectorId":1}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: DUAL_1
INFO:ocpp:CP_TEST: send [3,"e5125a08-f864-4524-89f8-553942875273",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"3ec25e51-d20e-45df-b780-c950287013f6","StartTransaction",{"connectorId":1,"idTag":"DUAL_1","meterStart":0,"timestamp":"2026-01-05T15:27:16.930333"}]
INFO:ocpp:CP_TEST: receive message [3,"e5125a08-f864-4524-89f8-553942875273",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"3ec25e51-d20e-45df-b780-c950287013f6","StartTransaction",{"connectorId":1,"idTag":"DUAL_1","meterStart":0,"timestamp":"2026-01-05T15:27:16.930333"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'DUAL_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:16.930333'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:MockSystemTest:8.1.3 RemoteStart (Conn 1): Accepted
INFO:ocpp:CP_TEST: send [3,"3ec25e51-d20e-45df-b780-c950287013f6",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"cd086d02-2cd9-4a42-af84-80ea700a295d","RemoteStartTransaction",{"idTag":"DUAL_2","connectorId":2}]
INFO:ocpp:CP_TEST: receive message [3,"3ec25e51-d20e-45df-b780-c950287013f6",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [2,"cd086d02-2cd9-4a42-af84-80ea700a295d","RemoteStartTransaction",{"idTag":"DUAL_2","connectorId":2}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: DUAL_2
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: send [3,"cd086d02-2cd9-4a42-af84-80ea700a295d",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"de359a3e-98d6-494d-a991-025fa68bbdb","StartTransaction",{"connectorId":2,"idTag":"DUAL_2","meterStart":0,"timestamp":"2026-01-05T15:27:16.932854"}]
INFO:ocpp:CP_TEST: receive message [3,"cd086d02-2cd9-4a42-af84-80ea700a295d",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"de359a3e-98d6-494d-a991-025fa68bbdb","StartTransaction",{"connectorId":2,"idTag":"DUAL_2","meterStart":0,"timestamp":"2026-01-05T15:27:16.932854"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 2, 'id_tag': 'DUAL_2', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:16.932854'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:MockSystemTest:8.1.4 RemoteStart (Conn 2): Accepted
WARNING:ocpp_16_interface:Transaction already in progress for connector 1.
INFO:MockSystemTest:8.1.5 StartTx (Conn 1): Sent
INFO:ocpp:CP_TEST: send [3,"de359a3e-98d6-494d-a991-025fa68bbdb",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"de359a3e-98d6-494d-a991-025fa68bbdb",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"b64dd168-229d-4f72-8a2f-57576f147e7b","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp_16_interface:Transaction started on 2: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2,"b64dd168-229d-4f72-8a2f-57576f147e7b","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"b64dd168-229d-4f72-8a2f-57576f147e7b",{}]
INFO:ocpp:CP_TEST: receive message [3,"b64dd168-229d-4f72-8a2f-57576f147e7b",{}]
INFO:MockSystemTest:8.1.6 Status (Conn 1): Charging
WARNING:ocpp_16_interface:Transaction already in progress for connector 2.
INFO:MockSystemTest:8.1.7 StartTx (Conn 2): Sent
INFO:ocpp:CP_TEST: send [2,"3f66ee9f-b1fe-4c4a-8201-676d781591e8","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"3f66ee9f-b1fe-4c4a-8201-676d781591e8","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"3f66ee9f-b1fe-4c4a-8201-676d781591e8",{}]
INFO:ocpp:CP_TEST: receive message [3,"3f66ee9f-b1fe-4c4a-8201-676d781591e8",{}]
INFO:MockSystemTest:8.1.8 Status (Conn 2): Charging
INFO:MockSystemTest:8.1.9 MeterValues (Conn 1): Verifying receipt
INFO:MockSystemTest:8.1.10 MeterValues (Conn 2): Verifying receipt
INFO:ocpp:CP_TEST: send [2,"e2a2f3f6-5626-43fb-a22f-e9736c38207b","RemoteStopTransaction",{"transactionId":123}]
INFO:ocpp:CP_TEST: receive message [2,"e2a2f3f6-5626-43fb-a22f-e9736c38207b","RemoteStopTransaction",{"transactionId":123}]
INFO:ocpp_16_interface:Received RemoteStopTransaction for transaction_id: 123
INFO:ocpp:CP_TEST: send [3,"e2a2f3f6-5626-43fb-a22f-e9736c38207b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"e2a2f3f6-5626-43fb-a22f-e9736c38207b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"75bbb66d-8159-4d95-aa4e-2568749cc3c9","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:18.942136","transactionId":123,"reason":"Remote"}]
INFO:MockSystemTest:8.1.11 RemoteStop (Conn 1): Sent
INFO:ocpp:CP_TEST: receive message [2,"75bbb66d-8159-4d95-aa4e-2568749cc3c9","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:18.942136","transactionId":123,"reason":"Remote"}]
```

```
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:18.942136', 'transaction_id': 123, 'reason': 'Remote'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3, "75bbb66d-8159-4d95-aa4e-2568749cc3c9", {"idTagInfo": {"status": "Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3, "75bbb66d-8159-4d95-aa4e-2568749cc3c9", {"idTagInfo": {"status": "Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:ocpp:CP_TEST: send [2, "5dca3986-3755-44cf-923e-7814af230849", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Finishing"}]
INFO:ocpp:CP_TEST: receive message [2, "5dca3986-3755-44cf-923e-7814af230849", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Finishing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "5dca3986-3755-44cf-923e-7814af230849", {}]
INFO:ocpp:CP_TEST: receive message [3, "5dca3986-3755-44cf-923e-7814af230849", {}]
INFO:MockSystemTest:8.1.12 StopTx (Conn 1): Completed
INFO:ocpp:CP_TEST: send [2, "138a670d-7ae3-40f2-8fb6-3bb09e3ddc9d", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:ocpp:CP_TEST: receive message [2, "138a670d-7ae3-40f2-8fb6-3bb09e3ddc9d", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "138a670d-7ae3-40f2-8fb6-3bb09e3ddc9d", {}]
INFO:ocpp:CP_TEST: receive message [3, "138a670d-7ae3-40f2-8fb6-3bb09e3ddc9d", {}]
INFO:MockSystemTest:8.1.14 Status (Conn 1): Available
INFO:ocpp:CP_TEST: send [2, "b61717c3-b98f-4063-9ac0-4e599ee56ea6", "RemoteStopTransaction", {"transactionId": 456}]
INFO:ocpp:CP_TEST: receive message [2, "b61717c3-b98f-4063-9ac0-4e599ee56ea6", "RemoteStopTransaction", {"transactionId": 456}]
INFO:ocpp_16_interface:Received RemoteStopTransaction for transaction_id: 456
INFO:ocpp:CP_TEST: send [3, "b61717c3-b98f-4063-9ac0-4e599ee56ea6", {"status": "Rejected"}]
INFO:ocpp:CP_TEST: receive message [3, "b61717c3-b98f-4063-9ac0-4e599ee56ea6", {"status": "Rejected"}]
INFO:MockSystemTest:8.1.15 RemoteStop (Conn 2): Sent
INFO:ocpp:CP_TEST: send [2, "374cf4a0-a350-4c59-abc9-0d3411576f4b", "StatusNotification", {"connectorId": 2, "errorCode": "NoError", "status": "Finishing"}]
INFO:ocpp:CP_TEST: receive message [2, "374cf4a0-a350-4c59-abc9-0d3411576f4b", "StatusNotification", {"connectorId": 2, "errorCode": "NoError", "status": "Finishing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Finishing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "374cf4a0-a350-4c59-abc9-0d3411576f4b", {}]
INFO:ocpp:CP_TEST: receive message [3, "374cf4a0-a350-4c59-abc9-0d3411576f4b", {}]
INFO:MockSystemTest:8.1.16 StopTx (Conn 2): Completed
INFO:ocpp:CP_TEST: send [2, "2d63c018-5f04-4e74-88ac-94c3a7f6a599", "StatusNotification", {"connectorId": 2, "errorCode": "NoError", "status": "Available"}]
INFO:ocpp:CP_TEST: receive message [2, "2d63c018-5f04-4e74-88ac-94c3a7f6a599", "StatusNotification", {"connectorId": 2, "errorCode": "NoError", "status": "Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "2d63c018-5f04-4e74-88ac-94c3a7f6a599", {}]
INFO:ocpp:CP_TEST: receive message [3, "2d63c018-5f04-4e74-88ac-94c3a7f6a599", {}]
INFO:MockSystemTest:8.1.18 Status (Conn 2): Available
INFO:MockSystemTest:
--- 8.2 Shared E-Stop ---
INFO:ocpp:CP_TEST: send [2, "10c2ebf4-6c69-49a2-afd8-a8a1e4be9aae", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:ocpp:CP_TEST: receive message [2, "10c2ebf4-6c69-49a2-afd8-a8a1e4be9aae", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "10c2ebf4-6c69-49a2-afd8-a8a1e4be9aae", {}]
INFO:ocpp:CP_TEST: receive message [3, "10c2ebf4-6c69-49a2-afd8-a8a1e4be9aae", {}]
INFO:ocpp:CP_TEST: send [2, "f2a514fa-edaf-4440-a0f5-e89a7a2f2ef9", "StatusNotification", {"connectorId": 2, "errorCode": "NoError", "status": "Available"}]
INFO:ocpp:CP_TEST: receive message [2, "f2a514fa-edaf-4440-a0f5-e89a7a2f2ef9", "StatusNotification", {"connectorId": 2, "errorCode": "NoError", "status": "Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "f2a514fa-edaf-4440-a0f5-e89a7a2f2ef9", {}]
INFO:ocpp:CP_TEST: receive message [3, "f2a514fa-edaf-4440-a0f5-e89a7a2f2ef9", {}]
INFO:ocpp:CP_TEST: send [2, "6154aa2d-abf1-4baa-b712-3d8c0f6ca552", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Preparing"}]
INFO:ocpp:CP_TEST: receive message [2, "6154aa2d-abf1-4baa-b712-3d8c0f6ca552", "StatusNotification", {"connectorId": 1, "errorCode": "NoError", "status": "Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "6154aa2d-abf1-4baa-b712-3d8c0f6ca552", {}]
INFO:ocpp:CP_TEST: receive message [3, "6154aa2d-abf1-4baa-b712-3d8c0f6ca552", {}]
INFO:MockSystemTest:8.2.2 Conn 1 Plug: Preparing
```



```
INFO:ocpp:CP_TEST: send [2,"3fa49e87-d52e-416e-9fc2-d71142f5d30b","RemoteStartTransaction",{"idTag":"ESTOP_1","connectorId":1}]
INFO:ocpp:CP_TEST: receive message [2,"3fa49e87-d52e-416e-9fc2-d71142f5d30b","RemoteStartTransaction",{"idTag":"ESTOP_1","connectorId":1}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: ESTOP_1
INFO:ocpp:CP_TEST: send [3,"3fa49e87-d52e-416e-9fc2-d71142f5d30b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"77baceca-eff2-4ca5-9529-b54e3ef8d2a3","StartTransaction",{"connectorId":1,"idTag":"ESTOP_1","meterStart":0,"timestamp":"2026-01-05T15:27:20.960232"}]
INFO:ocpp:CP_TEST: receive message [3,"3fa49e87-d52e-416e-9fc2-d71142f5d30b",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"77baceca-eff2-4ca5-9529-b54e3ef8d2a3","StartTransaction",{"connectorId":1,"idTag":"ESTOP_1","meterStart":0,"timestamp":"2026-01-05T15:27:20.960232"}]
INFO:MockSystemTest:8.2.3 RemoteStart (Conn 1): Accepted
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'ESTOP_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:20.960232'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"77baceca-eff2-4ca5-9529-b54e3ef8d2a3",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"77baceca-eff2-4ca5-9529-b54e3ef8d2a3",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: send [2,"31d3563a-4640-4c9d-a958-0d1fb4e0f8c7","StartTransaction",{"connectorId":1,"idTag":"ESTOP_1","meterStart":0,"timestamp":"2026-01-05T15:27:20.961574"}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2,"31d3563a-4640-4c9d-a958-0d1fb4e0f8c7","StartTransaction",{"connectorId":1,"idTag":"ESTOP_1","meterStart":0,"timestamp":"2026-01-05T15:27:20.961574"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'ESTOP_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:20.961574'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"31d3563a-4640-4c9d-a958-0d1fb4e0f8c7",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"31d3563a-4640-4c9d-a958-0d1fb4e0f8c7",{"transactionId":123,"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:8.2.4 StartTx (Conn 1): Sent
INFO:ocpp:CP_TEST: send [2,"5e38f03f-a17f-4b44-8d03-a5c1d1b0039a","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"5e38f03f-a17f-4b44-8d03-a5c1d1b0039a","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"5e38f03f-a17f-4b44-8d03-a5c1d1b0039a",{}]
INFO:ocpp:CP_TEST: receive message [3,"5e38f03f-a17f-4b44-8d03-a5c1d1b0039a",{}]
INFO:ocpp:CP_TEST: send [2,"7bf88288-d1d7-4de2-af72-4126a8d9adbc","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Preparing"}]
INFO:ocpp:CP_TEST: receive message [2,"7bf88288-d1d7-4de2-af72-4126a8d9adbc","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"7bf88288-d1d7-4de2-af72-4126a8d9adbc",{}]
INFO:ocpp:CP_TEST: receive message [3,"7bf88288-d1d7-4de2-af72-4126a8d9adbc",{}]
INFO:MockSystemTest:8.2.7 Conn 2 Plug: Preparing
INFO:ocpp:CP_TEST: send [2,"53a125ac-4272-49cc-9198-cec5f91ce409","RemoteStartTransaction",{"idTag":"ESTOP_2","connectorId":2}]
INFO:ocpp:CP_TEST: receive message [2,"53a125ac-4272-49cc-9198-cec5f91ce409","RemoteStartTransaction",{"idTag":"ESTOP_2","connectorId":2}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: ESTOP_2
INFO:ocpp:CP_TEST: send [3,"53a125ac-4272-49cc-9198-cec5f91ce409",{"status":"Rejected"}]
INFO:ocpp:CP_TEST: receive message [3,"53a125ac-4272-49cc-9198-cec5f91ce409",{"status":"Rejected"}]
INFO:MockSystemTest:8.2.8 RemoteStart (Conn 2): Rejected
WARNING:ocpp_16_interface:Transaction already in progress for connector 2.
INFO:MockSystemTest:8.2.9 StartTx (Conn 2): Sent
INFO:ocpp:CP_TEST: send [2,"04ad8248-4da7-4511-bbaa-0db954212a08","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Charging"}]
INFO:ocpp:CP_TEST: receive message [2,"04ad8248-4da7-4511-bbaa-0db954212a08","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"04ad8248-4da7-4511-bbaa-0db954212a08",{}]
INFO:ocpp:CP_TEST: receive message [3,"04ad8248-4da7-4511-bbaa-0db954212a08",{}]
INFO:ocpp:CP_TEST: send [2,"1dba549f-f97c-4c77-8270-bd48f2aaff9b","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:22.975050","transactionId":123,"reason":"EmergencyStop"}]
INFO:ocpp:CP_TEST: receive message [2,"1dba549f-f97c-4c77-8270-bd48f2aaff9b","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:22.975050","transactionId":123,"reason":"EmergencyStop"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:22.975050', 'transaction_id': 123, 'reason': 'EmergencyStop'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"1dba549f-f97c-4c77-8270-bd48f2aaff9b",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"1dba549f-f97c-4c77-8270-bd48f2aaff9b",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:8.2.12 StopTx (Conn 1 Emergency): Sent
INFO:ocpp:CP_TEST: send [2,"aaa714a5-9cf2-42da-a714-dfca5cd301d5","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:22.977779","transactionId":123,"reason":"EmergencyStop"}]
INFO:ocpp:CP_TEST: receive message [2,"aaa714a5-9cf2-42da-a714-dfca5cd301d5","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:22.977779","transactionId":123,"reason":"EmergencyStop"}]
```

```
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:22.977779', 'transaction_id': 123, 'reason': 'EmergencyStop'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"aaa714a5-9cf2-42da-a714-dfca5cd301d5",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"aaa714a5-9cf2-42da-a714-dfca5cd301d5",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 2: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:8.2.13 StopTx (Conn 2 Emergency): Sent
INFO:ocpp:CP_TEST: send [2,"1b7acdd4-7622-48d5-8475-94945cb48c64",{"StatusNotification":{"connectorId":1,"errorCode":"OtherError","status":"Faulted"}}]
INFO:ocpp:CP_TEST: receive message [2,"1b7acdd4-7622-48d5-8475-94945cb48c64",{"StatusNotification":{"connectorId":1,"errorCode":"OtherError","status":"Faulted"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'OtherError', 'status': 'Faulted'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"1b7acdd4-7622-48d5-8475-94945cb48c64",{}]
INFO:ocpp:CP_TEST: receive message [3,"1b7acdd4-7622-48d5-8475-94945cb48c64",{}]
INFO:ocpp:CP_TEST: send [2,"4815da7d-864d-4403-adc3-d887c77c319b",{"StatusNotification":{"connectorId":2,"errorCode":"OtherError","status":"Faulted"}}]
INFO:ocpp:CP_TEST: receive message [2,"4815da7d-864d-4403-adc3-d887c77c319b",{"StatusNotification":{"connectorId":2,"errorCode":"OtherError","status":"Faulted"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'OtherError', 'status': 'Faulted'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"4815da7d-864d-4403-adc3-d887c77c319b",{}]
INFO:ocpp:CP_TEST: receive message [3,"4815da7d-864d-4403-adc3-d887c77c319b",{}]
INFO:MockSystemTest:8.2.14 Status (Both): Faulted (Emergency)
INFO:ocpp:CP_TEST: send [2,"2e4455f8-5af8-4e6e-b01f-6cb879e30667",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"2e4455f8-5af8-4e6e-b01f-6cb879e30667",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"2e4455f8-5af8-4e6e-b01f-6cb879e30667",{}]
INFO:ocpp:CP_TEST: receive message [3,"2e4455f8-5af8-4e6e-b01f-6cb879e30667",{}]
INFO:ocpp:CP_TEST: send [2,"c21c1e32-0726-403d-bf32-fd65701dd75e",{"StatusNotification":{"connectorId":2,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"c21c1e32-0726-403d-bf32-fd65701dd75e",{"StatusNotification":{"connectorId":2,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"c21c1e32-0726-403d-bf32-fd65701dd75e",{}]
INFO:ocpp:CP_TEST: receive message [3,"c21c1e32-0726-403d-bf32-fd65701dd75e",{}]
INFO:MockSystemTest:8.2.16 Status (Both): Available (Recovered)
INFO:MockSystemTest:
--- 8.3 Power Loss (Dual) ---
INFO:ocpp:CP_TEST: send [2,"080f468b-9d69-44c3-a618-3163be098aba",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"080f468b-9d69-44c3-a618-3163be098aba",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"080f468b-9d69-44c3-a618-3163be098aba",{}]
INFO:ocpp:CP_TEST: receive message [3,"080f468b-9d69-44c3-a618-3163be098aba",{}]
INFO:ocpp:CP_TEST: send [2,"a569d7f8-3098-4da9-8cc5-24ae9ceebf68",{"StatusNotification":{"connectorId":2,"errorCode":"NoError","status":"Available"}}]
INFO:ocpp:CP_TEST: receive message [2,"a569d7f8-3098-4da9-8cc5-24ae9ceebf68",{"StatusNotification":{"connectorId":2,"errorCode":"NoError","status":"Available"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"a569d7f8-3098-4da9-8cc5-24ae9ceebf68",{}]
INFO:ocpp:CP_TEST: receive message [3,"a569d7f8-3098-4da9-8cc5-24ae9ceebf68",{}]
INFO:ocpp:CP_TEST: send [2,"f673ed76-42d8-44e4-b610-a7387c1f42fe",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Preparing"}}]
INFO:ocpp:CP_TEST: receive message [2,"f673ed76-42d8-44e4-b610-a7387c1f42fe",{"StatusNotification":{"connectorId":1,"errorCode":"NoError","status":"Preparing"}}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"f673ed76-42d8-44e4-b610-a7387c1f42fe",{}]
INFO:ocpp:CP_TEST: receive message [3,"f673ed76-42d8-44e4-b610-a7387c1f42fe",{}]
INFO:ocpp:CP_TEST: send [2,"f74c743e-36ed-4b7a-8d55-b24ecdd5bcec",{"RemoteStartTransaction":{"idTag":"PLOSS_1","connectorId":1}}]
INFO:ocpp:CP_TEST: receive message [2,"f74c743e-36ed-4b7a-8d55-b24ecdd5bcec",{"RemoteStartTransaction":{"idTag":"PLOSS_1","connectorId":1}}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: PLOSS_1
INFO:ocpp:CP_TEST: send [3,"f74c743e-36ed-4b7a-8d55-b24ecdd5bcec",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: send [2,"91c7bf0a-5e55-4805-b8bc-b21ac81de5b1",{"StartTransaction":{"connectorId":1,"idTag":"PLOSS_1","meterStart":0,"timestamp":"2026-01-05T15:27:22.994303"}}]
INFO:ocpp:CP_TEST: receive message [3,"f74c743e-36ed-4b7a-8d55-b24ecdd5bcec",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [2,"91c7bf0a-5e55-4805-b8bc-b21ac81de5b1",{"StartTransaction":{"connectorId":1,"idTag":"PLOSS_1","meterStart":0,"timestamp":"2026-01-05T15:27:22.994303"}}]
```

```
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'PLOSS_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:22.994303'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3, "91c7bf0a-5e55-4805-b8bc-b21ac81de5b1", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3, "91c7bf0a-5e55-4805-b8bc-b21ac81de5b1", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp:CP_TEST: send [2, "56566796-8fef-4468-9b97-c11961e79297", "StartTransaction", {"connectorId":1, "idTag": "PLOSS_1", "meterStart":0, "timestamp": "2026-01-05T15:27:22.996199"}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2, "56566796-8fef-4468-9b97-c11961e79297", "StartTransaction", {"connectorId":1, "idTag": "PLOSS_1", "meterStart":0, "timestamp": "2026-01-05T15:27:22.996199"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 1, 'id_tag': 'PLOSS_1', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:22.996199'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3, "56566796-8fef-4468-9b97-c11961e79297", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3, "56566796-8fef-4468-9b97-c11961e79297", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 1: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: send [2, "c626f332-b176-407b-b5cc-6207a087699f", "StatusNotification", {"connectorId":1, "errorCode": "NoError", "status": "Charging"}]
INFO:ocpp:CP_TEST: receive message [2, "c626f332-b176-407b-b5cc-6207a087699f", "StatusNotification", {"connectorId":1, "errorCode": "NoError", "status": "Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "c626f332-b176-407b-b5cc-6207a087699f", {}]
INFO:ocpp:CP_TEST: receive message [3, "c626f332-b176-407b-b5cc-6207a087699f", {}]
INFO:ocpp:CP_TEST: send [2, "ad18a757-adb9-4560-ba56-c480bd8e0024", "StatusNotification", {"connectorId":2, "errorCode": "NoError", "status": "Preparing"}]
INFO:ocpp:CP_TEST: receive message [2, "ad18a757-adb9-4560-ba56-c480bd8e0024", "StatusNotification", {"connectorId":2, "errorCode": "NoError", "status": "Preparing"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Preparing'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "ad18a757-adb9-4560-ba56-c480bd8e0024", {}]
INFO:ocpp:CP_TEST: receive message [3, "ad18a757-adb9-4560-ba56-c480bd8e0024", {}]
INFO:ocpp:CP_TEST: send [2, "3e465b31-84ea-42d8-90bd-50ba4d46fb3c", "RemoteStartTransaction", {"idTag": "PLOSS_2", "connectorId":2}]
INFO:ocpp:CP_TEST: receive message [2, "3e465b31-84ea-42d8-90bd-50ba4d46fb3c", "RemoteStartTransaction", {"idTag": "PLOSS_2", "connectorId":2}]
INFO:ocpp_16_interface:Received RemoteStartTransaction for id_tag: PLOSS_2
INFO:ocpp:CP_TEST: send [3, "3e465b31-84ea-42d8-90bd-50ba4d46fb3c", {"status": "Accepted"}]
INFO:ocpp:CP_TEST: send [2, "e2ce567f-a003-46cd-9fe4-b4f36159553c", "StartTransaction", {"connectorId":2, "idTag": "PLOSS_2", "meterStart":0, "timestamp": "2026-01-05T15:27:23.002924"}]
INFO:ocpp:CP_TEST: receive message [3, "3e465b31-84ea-42d8-90bd-50ba4d46fb3c", {"status": "Accepted"}]
INFO:ocpp:CP_TEST: receive message [2, "e2ce567f-a003-46cd-9fe4-b4f36159553c", "StartTransaction", {"connectorId":2, "idTag": "PLOSS_2", "meterStart":0, "timestamp": "2026-01-05T15:27:23.002924"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 2, 'id_tag': 'PLOSS_2', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:23.002924'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3, "e2ce567f-a003-46cd-9fe4-b4f36159553c", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3, "e2ce567f-a003-46cd-9fe4-b4f36159553c", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp:CP_TEST: send [2, "6508f3fc-cb48-44cb-8466-976a7e8ab3d0", "StartTransaction", {"connectorId":2, "idTag": "PLOSS_2", "meterStart":0, "timestamp": "2026-01-05T15:27:23.004204"}]
INFO:ocpp_16_interface:Transaction started on 2: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: receive message [2, "6508f3fc-cb48-44cb-8466-976a7e8ab3d0", "StartTransaction", {"connectorId":2, "idTag": "PLOSS_2", "meterStart":0, "timestamp": "2026-01-05T15:27:23.004204"}]
INFO:MockSystemTest:[CSMS] <- StartTransaction: {'connector_id': 2, 'id_tag': 'PLOSS_2', 'meter_start': 0, 'timestamp': '2026-01-05T15:27:23.004204'}
INFO:MockSystemTest:[CSMS] -> StartTransactionConf: StartTransaction(transaction_id=123, id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3, "6508f3fc-cb48-44cb-8466-976a7e8ab3d0", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3, "6508f3fc-cb48-44cb-8466-976a7e8ab3d0", {"transactionId":123, "idTagInfo":{"status": "Accepted"}}]
INFO:ocpp_16_interface:Transaction started on 2: 123
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:ocpp:CP_TEST: send [2, "c2f21e50-b55c-44d2-bd55-07d7498092e9", "StatusNotification", {"connectorId":2, "errorCode": "NoError", "status": "Charging"}]
INFO:ocpp:CP_TEST: receive message [2, "c2f21e50-b55c-44d2-bd55-07d7498092e9", "StatusNotification", {"connectorId":2, "errorCode": "NoError", "status": "Charging"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Charging'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3, "c2f21e50-b55c-44d2-bd55-07d7498092e9", {}]
INFO:ocpp:CP_TEST: receive message [3, "c2f21e50-b55c-44d2-bd55-07d7498092e9", {}]
INFO:MockSystemTest:8.3 Sequence: Both Charging
INFO:ocpp:CP_TEST: send [2, "1954e6f3-a00a-4d46-a2a0-b0023db61234", "StopTransaction", {"meterStop":100, "timestamp": "2026-01-05T15:27:24.008510", "transactionId":123, "reason": "PowerLoss"}]
INFO:ocpp:CP_TEST: receive message [2, "1954e6f3-a00a-4d46-a2a0-b0023db61234", "StopTransaction", {"meterStop":100, "timestamp": "2026-01-05T15:27:24.008510", "transactionId":123, "reason": "PowerLoss"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:24.008510', 'transaction_id': 123, 'reason': 'PowerLoss'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3, "1954e6f3-a00a-4d46-a2a0-b0023db61234", {"idTagInfo":{"status": "Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3, "1954e6f3-a00a-4d46-a2a0-b0023db61234", {"idTagInfo":{"status": "Accepted"}}]
```

```
INFO:ocpp_16_interface:Transaction stopped on 1: 123
INFO:MockSystemTest:8.3.12 StopTx (Conn 1 PowerLoss): Sent
INFO:ocpp:CP_TEST: send [2,"fbee16902-0400-4cdd-944b-4ba9dcc825f5","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:24.010651","transactionId":123,"reason":"PowerLoss"}]
INFO:ocpp:CP_TEST: receive message [2,"fbee16902-0400-4cdd-944b-4ba9dcc825f5","StopTransaction",{"meterStop":100,"timestamp":"2026-01-05T15:27:24.010651","transactionId":123,"reason":"PowerLoss"}]
INFO:MockSystemTest:[CSMS] <- StopTransaction: {'meter_stop': 100, 'timestamp': '2026-01-05T15:27:24.010651', 'transaction_id': 123, 'reason': 'PowerLoss'}
INFO:MockSystemTest:[CSMS] -> StopTransactionConf: StopTransaction(id_tag_info={'status': <AuthorizationStatus.accepted: 'Accepted'>})
INFO:ocpp:CP_TEST: send [3,"fbee16902-0400-4cdd-944b-4ba9dcc825f5",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp:CP_TEST: receive message [3,"fbee16902-0400-4cdd-944b-4ba9dcc825f5",{"idTagInfo":{"status":"Accepted"}}]
INFO:ocpp_16_interface:Transaction stopped on 2: 123
INFO:MockSystemTest:HARDWARE: Charger stopped.
INFO:MockSystemTest:8.3.13 StopTx (Conn 2 PowerLoss): Sent
INFO:ocpp:CP_TEST: send [2,"7ebee1f2-5455-41df-8f8b-4e029e9e680e","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2,"7ebee1f2-5455-41df-8f8b-4e029e9e680e","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:27:24.013678', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3,"7ebee1f2-5455-41df-8f8b-4e029e9e680e",{"currentTime":"2026-01-05T15:27:24.013678","interval":10,"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"7ebee1f2-5455-41df-8f8b-4e029e9e680e",{"currentTime":"2026-01-05T15:27:24.013678","interval":10,"status":"Accepted"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:8.3.14 BootNotification: Sent
INFO:ocpp:CP_TEST: send [2,"2af67d90-8a36-4c61-8af6-8c711d7c3a70","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"2af67d90-8a36-4c61-8af6-8c711d7c3a70","StatusNotification",{"connectorId":1,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 1, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"2af67d90-8a36-4c61-8af6-8c711d7c3a70",{}]
INFO:ocpp:CP_TEST: receive message [3,"2af67d90-8a36-4c61-8af6-8c711d7c3a70",{}]
INFO:ocpp:CP_TEST: send [2,"27be0256-556f-445d-a7ac-823d22995281","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Available"}]
INFO:ocpp:CP_TEST: receive message [2,"27be0256-556f-445d-a7ac-823d22995281","StatusNotification",{"connectorId":2,"errorCode":"NoError","status":"Available"}]
INFO:MockSystemTest:[CSMS] <- StatusNotification: {'connector_id': 2, 'error_code': 'NoError', 'status': 'Available'}
INFO:MockSystemTest:[CSMS] -> StatusNotificationConf: {}
INFO:ocpp:CP_TEST: send [3,"27be0256-556f-445d-a7ac-823d22995281",{}]
INFO:ocpp:CP_TEST: receive message [3,"27be0256-556f-445d-a7ac-823d22995281",{}]
INFO:MockSystemTest:
--- 9. MEA Specific Configuration (Full Sequence) ---
INFO:ocpp:CP_TEST: send [2,"4b31a44f-9add-4f39-afbf-138c8b331784","ChangeConfiguration",{"key":"HeartbeatInterval","value":"3600"}]
INFO:ocpp:CP_TEST: receive message [2,"4b31a44f-9add-4f39-afbf-138c8b331784","ChangeConfiguration",{"key":"HeartbeatInterval","value":"3600"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for HeartbeatInterval to 3600
INFO:ocpp:CP_TEST: send [3,"4b31a44f-9add-4f39-afbf-138c8b331784",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"4b31a44f-9add-4f39-afbf-138c8b331784",{"status":"Accepted"}]
INFO:MockSystemTest:9.1 Heartbeat 3600: Accepted
INFO:ocpp:CP_TEST: send [2,"0591a524-e7d3-411a-b5b5-93f5629d4c43","ChangeConfiguration",{"key":"MeterValueSampleInterval","value":"60"}]
INFO:ocpp:CP_TEST: receive message [2,"0591a524-e7d3-411a-b5b5-93f5629d4c43","ChangeConfiguration",{"key":"MeterValueSampleInterval","value":"60"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for MeterValueSampleInterval to 60
INFO:ocpp:CP_TEST: send [3,"0591a524-e7d3-411a-b5b5-93f5629d4c43",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"0591a524-e7d3-411a-b5b5-93f5629d4c43",{"status":"Accepted"}]
INFO:MockSystemTest:9.2 MeterValueSampleInterval 60: Accepted
INFO:ocpp:CP_TEST: send [2,"8ab65182-a328-4dc8-9728-f53f779a2a2c","ChangeConfiguration",{"key":"MEA_V2G_PowerDemand","value":"-5000"}]
INFO:ocpp:CP_TEST: receive message [2,"8ab65182-a328-4dc8-9728-f53f779a2a2c","ChangeConfiguration",{"key":"MEA_V2G_PowerDemand","value":"-5000"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for MEA_V2G_PowerDemand to -5000
INFO:ocpp_16_interface:[V2G] Grid Demand: Discharging 5000 W
INFO:ocpp:CP_TEST: send [3,"8ab65182-a328-4dc8-9728-f53f779a2a2c",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"8ab65182-a328-4dc8-9728-f53f779a2a2c",{"status":"Accepted"}]
INFO:MockSystemTest:9.3 V2G PowerDemand -5000 (Discharge): Accepted
INFO:ocpp:CP_TEST: send [2,"5da04db6-4520-4126-afc4-8251cadbf8aa8","ChangeConfiguration",{"key":"MEA_V2G_PowerDemand","value":"2000"}]
INFO:ocpp:CP_TEST: receive message [2,"5da04db6-4520-4126-afc4-8251cadbf8aa8","ChangeConfiguration",{"key":"MEA_V2G_PowerDemand","value":"2000"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for MEA_V2G_PowerDemand to 2000
INFO:ocpp_16_interface:[V2G] Grid Supply: Charging 2000 W
INFO:ocpp:CP_TEST: send [3,"5da04db6-4520-4126-afc4-8251cadbf8aa8",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"5da04db6-4520-4126-afc4-8251cadbf8aa8",{"status":"Accepted"}]
INFO:MockSystemTest:9.3 V2G PowerDemand 2000 (Charge): Accepted
```

```
INFO:ocpp:CP_TEST: send [2,"64be176e-e86c-4575-8383-815af2ea4e07","ChangeConfiguration",{"key":"MEA_V2G_PowerDemand","value":"0"}]
INFO:ocpp:CP_TEST: receive message [2,"64be176e-e86c-4575-8383-815af2ea4e07","ChangeConfiguration",{"key":"MEA_V2G_PowerDemand","value":"0"}]
INFO:ocpp_16_interface:Received ChangeConfiguration for MEA_V2G_PowerDemand to 0
INFO:ocpp_16_interface:[V2G] Idle
INFO:ocpp:CP_TEST: send [3,"64be176e-e86c-4575-8383-815af2ea4e07",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"64be176e-e86c-4575-8383-815af2ea4e07",{"status":"Accepted"}]
INFO:MockSystemTest:9.3 V2G PowerDemand 0: Accepted
INFO:ocpp:CP_TEST: send [2,"aea50a05-26d9-4df8-b4e9-ddacf3fc71fd","GetDiagnostics",{"location":"ftp://example.com/diagnostics"}]
INFO:ocpp:CP_TEST: receive message [2,"aea50a05-26d9-4df8-b4e9-ddacf3fc71fd","GetDiagnostics",{"location":"ftp://example.com/diagnostics"}]
INFO:ocpp_16_interface:Received GetDiagnostics for ftp://example.com/diagnostics
INFO:ocpp:CP_TEST: send [3,"aea50a05-26d9-4df8-b4e9-ddacf3fc71fd",{"fileName":"diagnostics.log"}]
INFO:ocpp:CP_TEST: receive message [3,"aea50a05-26d9-4df8-b4e9-ddacf3fc71fd",{"fileName":"diagnostics.log"}]
INFO:MockSystemTest:10.19 GetDiagnostics: diagnostics.log
INFO:ocpp:CP_TEST: send [2,"d0b80c28-1a9f-4c9c-8a6d-0d9eae28a778","UpdateFirmware",{"location":"ftp://example.com/firmware","retrieveDate":"2026-01-01T00:00:00Z"}]
INFO:ocpp:CP_TEST: receive message [2,"d0b80c28-1a9f-4c9c-8a6d-0d9eae28a778","UpdateFirmware",{"location":"ftp://example.com/firmware","retrieveDate":"2026-01-01T00:00:00Z"}]
INFO:ocpp_16_interface:Received UpdateFirmware from ftp://example.com/firmware
INFO:ocpp:CP_TEST: send [3,"d0b80c28-1a9f-4c9c-8a6d-0d9eae28a778",{}]
INFO:ocpp:CP_TEST: receive message [3,"d0b80c28-1a9f-4c9c-8a6d-0d9eae28a778",{}]
INFO:MockSystemTest:10.20 UpdateFirmware: Sent
INFO:ocpp:CP_TEST: send [2,"9dc4ac51-573d-4267-8a21-b05a78741a8d","SendLocalList",{"listVersion":1,"updateType":"Full","localAuthorizationList":[{"idTag":"TAG1","idTagInfo":{"status":"Accepted"}}]}]
INFO:ocpp:CP_TEST: receive message [2,"9dc4ac51-573d-4267-8a21-b05a78741a8d","SendLocalList",{"listVersion":1,"updateType":"Full","localAuthorizationList":[{"idTag":"TAG1","idTagInfo":{"status":"Accepted"}}]}]
INFO:ocpp_16_interface:Received SendLocalList
INFO:ocpp:CP_TEST: send [3,"9dc4ac51-573d-4267-8a21-b05a78741a8d",{"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"9dc4ac51-573d-4267-8a21-b05a78741a8d",{"status":"Accepted"}]
INFO:MockSystemTest:10.21 SendLocalList: Accepted
INFO:ocpp:CP_TEST: send [2,"795828d2-5200-44b5-bfab-1f2cb64af46f","GetLocalListVersion",{}]
INFO:ocpp:CP_TEST: receive message [2,"795828d2-5200-44b5-bfab-1f2cb64af46f","GetLocalListVersion",{}]
INFO:ocpp_16_interface:Received GetLocalListVersion
INFO:ocpp:CP_TEST: send [3,"795828d2-5200-44b5-bfab-1f2cb64af46f",{"listVersion":1}]
INFO:ocpp:CP_TEST: receive message [3,"795828d2-5200-44b5-bfab-1f2cb64af46f",{"listVersion":1}]
INFO:MockSystemTest:10.22 GetLocalListVersion: 1
INFO:MockSystemTest:
--- 10. Summary Verification ---
INFO:MockSystemTest:10.1 Log verification: Implicit via this test execution log.
INFO:MockSystemTest:
--- 11. Other Verification ---
INFO:MockSystemTest:11.1 Simulating LAN Disconnect...
INFO:MockSystemTest:11.1 Reconnecting...
INFO:websockets.server:connection open
INFO:MockSystemTest:CSMS: Client connected! CP_TEST
INFO:ocpp:CP_TEST: send [2,"3ded74bf-5d7c-4076-92be-8e1555225cc4","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:ocpp:CP_TEST: receive message [2,"3ded74bf-5d7c-4076-92be-8e1555225cc4","BootNotification",{"chargePointModel":"MEA-V2G-01","chargePointVendor":"KMUTNB"}]
INFO:MockSystemTest:[CSMS] <- BootNotification: {'charge_point_model': 'MEA-V2G-01', 'charge_point_vendor': 'KMUTNB'}
INFO:MockSystemTest:[CSMS] -> BootNotificationConf: BootNotification(current_time='2026-01-05T15:27:26.040427', interval=10, status=<RegistrationStatus.accepted: 'Accepted'>)
INFO:ocpp:CP_TEST: send [3,"3ded74bf-5d7c-4076-92be-8e1555225cc4",{"currentTime":"2026-01-05T15:27:26.040427","interval":10,"status":"Accepted"}]
INFO:ocpp:CP_TEST: receive message [3,"3ded74bf-5d7c-4076-92be-8e1555225cc4",{"currentTime":"2026-01-05T15:27:26.040427","interval":10,"status":"Accepted"}]
INFO:ocpp_16_interface:Connected to central system.
INFO:MockSystemTest:HARDWARE: Charger started.
INFO:MockSystemTest:11.1 Reconnect BootNotification: Sent
INFO:MockSystemTest:Test Suite Completed.
ERROR:websockets.server:connection handler failed
Traceback (most recent call last):
  File "/Users/rus/Library/Python/3.12/lib/python/site-packages/websockets/asyncio/server.py", line 376, in conn_handler
    await self.handler(connection)
  File "/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py", line 143, in on_connect
    await cp.start()
  File "/Users/rus/Library/Python/3.12/lib/python/site-packages/ocpp/charge_point.py", line 246, in start
    message = await self._connection.recv()
    ~~~~~
  File "/Users/rus/Library/Python/3.12/lib/python/site-packages/websockets/asyncio/connection.py", line 322, in recv
```

```
        raise self.protocol.close_exc from self.recv_exc
websockets.exceptions.ConnectionClosedOK: received 1000 (OK); then sent 1000 (OK)
INFO:websockets.server:server closing
ERROR:websockets.server:connection handler failed
Traceback (most recent call last):
  File "/Users/rus/Library/Python/3.12/lib/python/site-packages/websockets/asyncio/server.py", line 376, in conn_handler
    await self.handler(connection)
  File "/Users/rus/development/MEA-V2G/tests/system/test_mea_mock.py", line 143, in on_connect
    await cp.start()
  File "/Users/rus/Library/Python/3.12/lib/python/site-packages/ocpp/charge_point.py", line 246, in start
    message = await self._connection.recv()
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/rus/Library/Python/3.12/lib/python/site-packages/websockets/asyncio/connection.py", line 322, in recv
    raise self.protocol.close_exc from self.recv_exc
websockets.exceptions.ConnectionClosedOK: sent 1001 (going away); then received 1001 (going away)
INFO:websockets.server:server closed
```